Constraints of Asymptotically Safe Quantum Gravity on Scalar-Tensor Theories and Cosmic Strings

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Abstract

This essay examines the constraints that the asymptotic-safety paradigm imposes on two prominent classes of phenomenological models: shift-symmetric Horndeski scalar-tensor theories proposed as explanations of dynamical dark energy, and Abelian Higgs models capable of producing cosmic strings. Using the Functional Renormalization Group (FRG) as the central analytical instrument, we demonstrate that the requirement of ultraviolet (UV) completeness through quantum scale symmetry severely restricts the viable infrared (IR) parameter space of both model classes. In the Horndeski sector, the cubic braiding coupling is shown to be symmetry-protected and vanishes identically at the shifted Gaussian fixed point, rendering kinetic braiding models incapable of generating the observed dark energy density. In the cosmic string sector, the Abelian Higgs model cannot accommodate spontaneous symmetry breaking at sub-Planckian scales without the introduction of at least forty-two additional Dirac fermions. We place these results in the broader context of the Hubble tension, the NANOGrav 15-year dataset, and the conceptual architecture of the asymptotically safe landscape, arguing that UV consistency acts as a structural filter that collapses the apparently large space of beyond-the-Standard-Model candidates into a much smaller set of genuinely viable theories.

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1 Introduction: The Crisis in Cosmology and the Need for UV Completion

Contemporary theoretical cosmology finds itself at an unusual impasse. The Standard Model of Cosmology, Λ-CDM, rests on two invisible pillars: cold dark matter, which provides the gravitational scaffolding for large-scale structure, and the cosmological constant Λ, a static energy density intrinsic to empty space. For several decades this framework provided excellent agreement with observation. As measurement precision has improved, however, significant internal tensions have emerged that resist accommodation within the Λ-CDM paradigm. The most comprehensive recent analysis of cosmological parameters from the Planck satellite [40] and the independent distance-ladder measurements assembled by Riess and collaborators [41] are in tension at more than 4σ, establishing what is now widely called the Hubble tension as a genuine crisis rather than a statistical fluctuation.

The most discussed manifestation of this crisis is the Hubble tension. Local measurements of the expansion rate H_0 via Cepheid variable stars and Type Ia supernovae consistently yield higher values than those inferred from the CMB assuming Λ-CDM cosmology [41, 40]. A second tension concerns the parameter σ_8, which characterizes the amplitude of matter clustering on large scales. CMB-derived predictions of σ_8 exceed those inferred from weak gravitational lensing surveys of the late-time universe, suggesting that Λ-CDM may overpredict the rate of structure growth [37, 38]. Neither tension constitutes a definitive falsification of Λ-CDM, but together they suggest the presence of physics beyond the static cosmological constant.

This situation has motivated an enormous literature on dynamical dark energy (DDE), much of it invoking scalar fields coupled to gravity through derivative interactions [35]. The most mathematically systematic framework for such theories is Horndeski gravity [33], the most general scalar-tensor theory producing second-order equations of motion in four spacetime dimensions. Within this class, kinetic braiding models are particularly widely studied: the Galileon and its relatives [34] mix the kinetic term of the scalar field with spacetime curvature in a way that can drive accelerated expansion without introducing a bare cosmological constant, and they reproduce the correct speed of gravitational waves $c_t = 1$ when the non-minimal coupling functions are appropriately restricted.

What is typically absent from such constructions is an account of their ultraviolet behavior. Phenomenological dark energy models are built as effective field theories valid at cosmological scales of order H_0^{-1}, with no guarantee of mathematical consistency at the Planck scale $M_{\rm Pl} \approx 10^{19}\,$GeV. This is not merely an aesthetic deficiency. A theory whose couplings diverge or become ill-defined at

finite energy is not a theory of nature; it is a placeholder for one [3, 5]. The requirement that a viable model admit a well-defined, interacting UV limit is therefore a genuine physical constraint, not a matter of theoretical preference.

The asymptotic-safety program provides precisely this constraint. Proposed originally by Weinberg [13] and realized concretely through the Functional Renormalization Group framework developed by Wetterich [9] and subsequently applied to gravity by Reuter [14], asymptotic safety posits that the renormalization group flow of quantum gravity approaches a non-Gaussian fixed point (NGFP) in the deep UV. At this fixed point, quantum fluctuations balance each other out in a condition of quantum scale symmetry. The theory remains interacting but finite, free of Landau poles and perturbative divergences. Crucially, the fixed-point structure imposes rigid constraints on the matter sector that must coexist with quantum gravity [17, 18]. Not every low-energy effective theory can be connected by a smooth renormalization group trajectory to an asymptotically safe UV fixed point. Those that cannot are, in the terminology introduced by Vafa and elaborated by subsequent researchers, swampland candidates—mathematically problematic extrapolations that cannot represent fundamental physics [49].

The present essay synthesizes recent results establishing these constraints for two model classes of direct phenomenological relevance. The first is the family of shift-symmetric Horndeski theories with kinetic braiding, which are among the most popular candidate explanations of dynamical dark energy [65]. The second is the Abelian Higgs model, the minimal field-theoretic mechanism for cosmic string production [47]. For each model class, we describe the FRG methodology, state the main theoretical results, and connect them to observational data, particularly the NANOGrav 15-year dataset [42] and the Hubble tension measurements.

The conclusion that emerges is what we term a small-landscape picture: the requirement of asymptotic safety collapses the apparently vast space of beyond-Standard-Model phenomenological models into a very small set of genuinely UV-consistent theories [60]. This is not a limitation of the paradigm but its most powerful feature.

2 The Asymptotic-Safety Paradigm: Fixed Points, Scaling, and Predictivity

2.1 The Ultraviolet Problem in Quantum Gravity

General Relativity is perturbatively non-renormalizable. Power-counting arguments show that Newton's coupling G has mass dimension -2 in four spacetime

dimensions, so higher-loop diagrams introduce operators of arbitrarily high dimension with independent coupling constants [7, 8]. To renormalize the theory order by order requires an infinite number of counterterms, which means an infinite number of free parameters that must be fixed by measurement. The theory loses predictive power at the Planck scale and above. This is the ultraviolet problem of quantum gravity.

Two broad strategies have been developed to address it. The first invokes new degrees of freedom at short distances—strings, loops, or higher-dimensional structures—that alter the theory's short-distance behavior by changing its fundamental constituents. The second, which is the approach underlying asymptotic safety [13], remains within the framework of quantum field theory and the metric, but seeks a non-trivial UV fixed point under the renormalization group. This fixed point plays the role that asymptotic freedom plays in non-Abelian gauge theories: it ensures that all physical couplings remain finite as the energy scale k is taken to infinity.

The key insight, due originally to Wilson and Kogut [2, 1] and formalized for gravity by Weinberg [13], is that perturbative non-renormalizability does not preclude the existence of a well-defined non-perturbative theory. A theory is asymptotically safe if the renormalization group flow, when expressed in terms of dimensionless couplings, has an interacting fixed point with a finite-dimensional UV-critical surface. The dimensionality of this critical surface determines the number of experimentally measurable parameters, and hence the predictive content of the theory [15, 16].

2.2 The Functional Renormalization Group and the Wetterich Equation

The primary technical tool of the asymptotic-safety program is the Functional Renormalization Group, specifically the exact flow equation for the scale-dependent effective action Γ_k derived by Wetterich [9] (see Appendix A for a derivation):

$$k\partial_k\Gamma_k = \frac{1}{2}\,\mathrm{Tr}\Big[\big(\Gamma_k^{(2)} + R_k\big)^{-1} k\partial_k R_k\Big]\,. \tag{1}$$

Here $\Gamma_k^{(2)}$ denotes the second functional derivative of the effective action with respect to the fields, evaluated at a background configuration, and R_k is an infrared regulator that suppresses the contribution of momentum modes below the sliding scale k. As k is lowered from a high initial scale Λ_{UV} toward zero, progressively more fluctuations are integrated out and Γ_k flows from a bare UV action to the full quantum effective action $\Gamma \equiv \Gamma_{k\to 0}$. Equation (1) has a one-loop structural form

but is exact: no approximations have been made in its derivation [10, 12]. All approximations enter when one truncates the infinite-dimensional theory space to a manageable finite-dimensional subspace—a physically motivated set of field operators whose couplings are tracked. The quality of the truncation determines the reliability of the resulting beta functions and fixed-point analysis [11].

A fixed point of the flow is a point $\{g_i^*\}$ in the space of dimensionless couplings where all beta functions vanish:

$$\beta_i(g^*) \equiv k\partial_k g_i\big|_{g=g^*} = 0. \tag{2}$$

The stability properties of the fixed point are encoded in the stability matrix $B_{ij} = \partial\beta_i/\partial g_j|_{g^*}$, whose eigenvalues θ_I are the critical exponents (see Appendix E). A coupling with $\mathrm{Re}(\theta_I) > 0$ is *relevant*: it can deviate from its fixed-point value as the scale flows to the IR, and must be fixed by experiment. A coupling with $\mathrm{Re}(\theta_I) < 0$ is *irrelevant*: it is driven back to its fixed-point value and therefore constitutes a prediction of the theory, not a free parameter. The number of relevant directions equals the number of free parameters of the asymptotically safe theory—a finite number, in sharp contrast to the infinite-parameter expansion of perturbative quantum gravity [20, 6].

2.3 The Non-Gaussian Fixed Point of Quantum Gravity

Evidence for the existence of a non-Gaussian fixed point (NGFP) in the Einstein-Hilbert truncation, where Γ_k is restricted to the Ricci scalar and cosmological constant, was established by Reuter [14] and has since been confirmed in progressively larger truncations by Reuter and Saueressig [15] and many subsequent works reviewed by Eichhorn [17], Percacci [16], and Eichhorn and Schiffer [18]. The NGFP is characterized by finite values of the dimensionless Newton coupling $g = Gk^2$ and the dimensionless cosmological constant $\lambda = \Lambda/k^2$, together with two relevant directions corresponding to deformations of these two parameters. The fixed-point values and critical exponents are regulator-dependent but qualitatively robust: there exists an interacting UV fixed point around which the theory is well-defined and predictive.

The inclusion of matter fields modifies the fixed-point structure. Scalar fields, fermions, and gauge bosons each contribute differently to the beta functions for the gravitational couplings, and the gravitational fluctuations in turn contribute to the beta functions for the matter couplings [45, 18]. It is this gravitational back-reaction on the matter sector that generates the constraints on phenomenological models that are the primary subject of the present essay.

2.4 The Weak-Gravity Bound

A central organizing concept for matter-gravity systems in the asymptotic-safety framework is the weak-gravity bound [18]. When gravitational fluctuations are included, interacting fixed points in the matter sector that exist in the absence of gravity may be displaced or destabilized. As the dimensionless Newton coupling g increases from zero, interacting fixed points in the matter sector move through theory space. Beyond a critical value g_{crit}, pairs of fixed points collide and annihilate into the complex plane, leaving no real fixed point. The matter theory thereby loses its UV completion.

Quantitatively, within the Einstein-Hilbert truncation and for $\lambda = 0$, this critical value is found to be approximately $g_{\text{crit}} \approx 1.4$. The precise number is truncation-dependent, but the qualitative mechanism is robust: strong gravitational fluctuations destroy the fixed-point structure of the matter sector [18, 17]. The weak-gravity bound therefore defines a forbidden region in the gravitational coupling plane and constrains both the values of the gravitational couplings at the UV fixed point and the matter content of any UV-complete gravity-matter system.

3 Theory Space and the Architecture of RG Flow

3.1 Theory Space as a Dynamical System

To formulate the asymptotic-safety condition with mathematical precision, it is useful to introduce the notion of theory space. Theory space $\mathcal{T}$ is the space of all action functionals consistent with the field content and symmetries under consideration [2, 3]. Each point in $\mathcal{T}$ corresponds to a specific effective action

$$\Gamma = \sum_i g_i \mathcal{O}_i, \tag{3}$$

where $\mathcal{O}_i$ denote operators constructed from the fields and their derivatives, and g_i are the associated coupling constants.

The renormalization group defines a flow on $\mathcal{T}$:

$$k\partial_k g_i = \beta_i(g_1, g_2, \ldots). \tag{4}$$

The set of beta functions therefore defines a vector field on theory space, and the RG flow corresponds to the integral curves of this vector field [1, 6, 20]. The RG flow can equally be interpreted as a trajectory in the coupling space, moving from the UV regime toward the IR as the resolution scale k is lowered. This interpretation, originating in the block-spin approach of Kadanoff [21] and Wilson's subse-

quent formalization [1, 2], reveals the renormalization group as a coarse-graining operation on the physical degrees of freedom: each step in the flow integrates out a thin shell of momentum modes and produces a new, equivalent description at a lower resolution.

The UV-critical surface is the subset of $\mathcal{T}$ consisting of all RG trajectories that reach the UV fixed point as $k \to \infty$. Its codimension equals the number of irrelevant directions at the fixed point. Physical theories are those whose RG trajectories lie on the UV-critical surface, since only they possess a UV-complete definition. All other points in $\mathcal{T}$ correspond to effective field theories that eventually lose predictive power at the energy scale where the couplings diverge.

3.2 Canonical Scaling and Dimensionless Couplings

Renormalization group analysis is most naturally expressed in terms of dimensionless couplings [7, 8, 5]. If a coupling g_i has canonical mass dimension d_i, we define the dimensionless quantity

$$\tilde{g}_i = k^{-d_i} g_i. \tag{5}$$

The RG equation for $\tilde{g}_i$ then takes the form

$$k\partial_k \tilde{g}_i = -d_i \tilde{g}_i + \beta_i^{\text{quantum}}(\tilde{g}). \tag{6}$$

The first term represents canonical scaling arising from dimensional analysis, while the second term encodes quantum corrections generated by fluctuations [10, 12].

For the Horndeski truncation considered in this work, the cubic braiding coupling h has canonical dimension -3 and the quartic coupling g has canonical dimension -4 in four spacetime dimensions. Both interactions are therefore power-counting irrelevant in perturbation theory, with canonical scaling terms $+3\tilde{h}$ and $+4\tilde{g}$ in equation (6). Their fate in the ultraviolet is determined by the interplay between this canonical suppression and quantum corrections from gravitational fluctuations [65]. In the absence of gravity the canonical scaling dominates and both couplings are irrelevant; the question is whether gravity introduces sufficient antiscreening to promote either coupling to relevance. The answer, as we shall see, is no.

3.3 Information Geometry of Coupling Space

The space of couplings $\{g_i\}$ itself carries a natural geometric structure when equipped with a metric [23, 24, 25]. The Fisher information matrix

$$G_{ij}(\theta) = \mathbb{E}[\partial_i \log P \; \partial_j \log P] \tag{7}$$

defines a Riemannian metric on the parameter space of a statistical model. In the renormalization group context, this idea motivates an information-geometric interpretation of RG trajectories as geodesics on the manifold of coupling constants, where the metric measures the distinguishability between nearby effective theories [26]. Fixed points are then critical points of an entropy-like functional on theory space, and the stability matrix B_{ij} encodes the local curvature of this landscape.

While the information-geometric perspective does not alter the technical content of the FRG analysis, it provides a clarifying conceptual framework. The asymptotically safe fixed point is a state of maximal self-consistency: the effective theory at every scale is identical, up to a rescaling of units, to the theory at every other scale. This is the field-theoretic realization of scale symmetry, and the Fisher information geometry makes precise the sense in which all UV-complete theories are parametrically close to one another in the vicinity of the fixed point [25].

3.4 Symmetry Constraints on Renormalization Group Flows

Symmetries of the effective action impose powerful constraints on the structure of beta functions, and this constraint is responsible for some of the most robust predictions in the asymptotic-safety literature [17]. Suppose the action is invariant under a discrete symmetry transformation $\Phi \to \mathcal{S}\Phi$. Operators in the action may transform either trivially or with a sign under this symmetry. If an operator $\mathcal{O}_i$ is odd under the symmetry while all other operators are even, then the corresponding beta function must take the form

$$\beta_i = g_i \, f(g_j), \tag{8}$$

for some function f of the remaining couplings. The coupling g_i therefore factors out of its own beta function. Consequently, $g_i = 0$ is always a fixed point of the RG flow. Such a coupling is *symmetry-protected*: quantum fluctuations cannot generate it radiatively if it vanishes in the ultraviolet [65].

This mechanism is far more general than the specific Horndeski application considered here. It applies whenever a theory space contains operators of mixed

parity under a preserved symmetry, and its consequences are exact within any truncation that respects the symmetry. The practical implication is that symmetry analysis, rather than detailed numerical computation, can often determine the fate of a coupling at the fixed point [18].

4 Shift-Symmetric Horndeski Gravity and the Asymptotic-Safety Constraint

4.1 Horndeski Theories and Kinetic Braiding

Horndeski gravity [33] identifies the most general scalar-tensor theory in four dimensions whose equations of motion remain second-order in derivatives. This property prevents the introduction of additional propagating degrees of freedom with the wrong sign kinetic term (Ostrogradski ghosts), making Horndeski theories the preferred framework for stable dynamical dark energy models [35, 34]. Within this landscape, shift-symmetric theories occupy a special position. Shift symmetry is the invariance of the action under the constant shift $\phi \to \phi + a$. This symmetry restricts all interactions to involve only derivatives of ϕ, and the fundamental kinetic variable is

$$\chi = -\frac{1}{2}\nabla_\mu\phi\,\nabla^\mu\phi. \tag{9}$$

The kinetic braiding sector studied here includes three operators beyond the Einstein-Hilbert term: the standard kinetic term χ, the cubic braiding term $h\chi\Box\phi$, and a quartic interaction $g\chi^2$. The coupling h parametrizes the braiding between the scalar field's kinetic term and the metric, giving rise to the braiding parameter α_B which appears in modified gravitational slip and deviations of the effective dark energy equation of state from -1. Observational constraints from the simultaneous detection of gravitational waves and their electromagnetic counterpart GW170817 established that the speed of tensor modes must satisfy $|c_t - 1| \lesssim 10^{-15}$ [43], which forces $G_{4,X} = 0$ and $G_5 = 0$ within the Horndeski framework and leaves the cubic braiding term as the primary candidate mechanism for kinetic dark energy.

4.2 FRG Truncation for Horndeski Models

To analyze the asymptotic-safety constraints on these models, one truncates the effective action as

$$\Gamma_k = \int d^4x\,\sqrt{g}\left[-\frac{M_{\rm Pl}^2}{2}(R-2\Lambda) + \chi + h\chi\Box\phi + g\chi^2 + \zeta\chi R\right]. \tag{10}$$

The gravitational sector is truncated to the Einstein-Hilbert form, with dimensionless Newton coupling G and dimensionless cosmological constant Λ. The gauge-fixing is performed in the Landau-DeWitt gauge $\alpha = 0$, $\beta = 0$, which simplifies the propagator structure and enables analytic evaluation of momentum traces. The metric fluctuations are decomposed into irreducible components using the York decomposition (Appendix B), separating the spin-2 transverse-traceless modes, vector modes, and scalar trace modes. The Litim regulator

$$R_k(p^2) = (k^2 - p^2)\,\Theta(k^2 - p^2) \tag{11}$$

is chosen for its analytic tractability with the Heaviside step function Θ, which reduces momentum integrals in the trace to simple rational functions of k [10, 12]. The evaluation of functional traces uses heat-kernel techniques summarized in Appendix C.

4.3 The Shifted Gaussian Fixed Point and the Vanishing of Braiding

The central result of the FRG analysis of shift-symmetric Horndeski models is the identification of the shifted Gaussian fixed point (SGFP). This fixed point inherits its name from the fact that, in the absence of gravity, the couplings h and g both have a Gaussian fixed point at zero; the inclusion of gravitational fluctuations shifts the fixed-point value of g while leaving that of h at zero [65].

The vanishing of h at the fixed point is not accidental; it is a consequence of the symmetry structure described in Section 3.4. The operator $\chi\Box\phi$ transforms with a definite sign under the discrete $\mathbb{Z}_2$ transformation $\phi \to -\phi$ while χ is invariant: the cubic term breaks this kinetic $\mathbb{Z}_2$ symmetry. Because the remaining operators in the truncation preserve this symmetry, the beta function for h must take the form

$$\beta_h = h\, f(G, \Lambda, g) + \mathcal{O}(h^3), \tag{12}$$

for some function f of the remaining couplings. This means that $h = 0$ is always a solution of $\beta_h = 0$, independently of the values of the gravitational couplings. Quantum gravity fluctuations cannot generate the braiding coupling radiatively from zero. The SGFP therefore satisfies $h_* = 0$ exactly [65].

The critical exponent associated with h at the SGFP is negative, confirming that h is an irrelevant coupling. Its infrared value is predicted by the RG flow: it remains zero at all scales below the Planck scale, driven to its fixed-point value by the RG evolution. The braiding parameter α_B vanishes identically. Even with the inclusion of the non-minimal coupling $\zeta\chi R$, the sign of the critical exponent for h

does not change, reinforcing the robustness of the conclusion [65, 18].

4.4 Phenomenological Failure and the Magnitude Mismatch

The prediction $h_* = 0$ creates a fundamental conflict with the phenomenological requirements of dynamical dark energy. To replace the cosmological constant with a dynamical scalar field whose energy density matches the observed value $\rho_\Lambda \approx (2 \times 10^{-3}\,\text{eV})^4$, the coupling g must take the value

$$g_{\text{required}} \approx \frac{1}{H_0^2 M_{\text{Pl}}^2} \approx 9 \times 10^7\,\text{eV}^{-4}. \tag{13}$$

The RG analysis within the EFT framework yields an upper bound on the infrared value of g of order

$$g_{\text{predicted}} \lesssim \frac{H^2}{M_{\text{Pl}}^2} \approx 10^{-7}\,\text{eV}^{-4}, \tag{14}$$

a discrepancy of approximately fifty-seven orders of magnitude [65]. When $h = 0$ is substituted into the expression for the scalar field energy density in the kinetic braiding model, one obtains

$$\rho_\phi = -\frac{\dot{\phi}^2}{4} < 0, \tag{15}$$

which is negative—contradicting the basic requirement of a positive total energy density. The shift-symmetric kinetic braiding model, precisely in the limit imposed by asymptotic safety, yields a scalar field that contributes negatively to the energy budget of the universe.

4.5 Effective Cosmological Dynamics

To connect the RG analysis directly to cosmological observables, consider the scalar-tensor system on a spatially flat Friedmann-Lemaître-Robertson-Walker background:

$$ds^2 = -dt^2 + a(t)^2 d\vec{x}^2. \tag{16}$$

The Friedmann equation takes the form

$$H^2 = \frac{1}{3M_{\text{Pl}}^2}\left(\rho_m + \rho_\phi\right), \tag{17}$$

where $H = \dot{a}/a$ is the Hubble parameter and ρ_m is the matter density [37, 39]. For the shift-symmetric scalar sector, the energy density is

$$\rho_\phi = \dot{\phi}^2 G_{2,\chi} - G_2 + 3H\dot{\phi}^3\, G_{3,\chi}. \tag{18}$$

The term proportional to $G_{3,\chi}$ corresponds to kinetic braiding and generates deviations from the cosmological constant equation of state. If the RG flow enforces $h = 0$, this contribution vanishes identically, and the scalar field reduces to a minimally coupled kinetic component with $\rho_\phi = -\dot{\phi}^2/4$. The renormalization group therefore constrains cosmological dynamics by fixing the coefficients entering the background equations [65, 36]. The absence of the braiding term implies that the scalar sector cannot produce the dynamical dark energy behavior required to resolve the Hubble tension within the shift-symmetric Horndeski framework.

4.6 Implications for the Hubble Tension

The Hubble tension has stimulated proposals for dynamical dark energy as a possible resolution: if the equation of state of dark energy departed from $w = -1$ in the recent past, the inferred value of H_0 from the CMB might shift upward toward the locally measured value [40, 41]. Shift-symmetric kinetic braiding models are natural candidates for such a scenario because they can in principle generate $w \neq -1$ and a non-trivial gravitational slip.

The asymptotic-safety constraint eliminates this possibility at the level of fundamental theory. If the braiding coupling h is zero at all energy scales, the scalar field cannot produce the kinetic dark energy required to generate a departure from $w = -1$ [65]. Any resolution of the Hubble tension through shift-symmetric Horndeski dynamics must therefore invoke either non-polynomial truncations of the scalar potential, extensions beyond the shift-symmetric sector, or a different class of scalar-tensor theories entirely.

5 Cosmic Strings in the Asymptotically Safe Landscape

5.1 Cosmic Strings and the Abelian Higgs Model

Cosmic strings are one-dimensional topological defects formed during phase transitions in the early universe when a continuous symmetry is spontaneously broken [36]. In the simplest realization they arise from the Abelian Higgs model: a $U(1)$ gauge theory with a charged complex scalar ϕ whose potential admits a circle of vacua at $|\phi| = v$. When the universe cools through the phase transition temperature, the scalar field acquires a non-zero vacuum expectation value (VEV), and topologically stable string-like configurations are formed at the boundaries between regions that select different phases. The tension of such strings is set by the symmetry-breaking scale v, and observational constraints from pulsar timing arrays bound $v \lesssim 4.6 \times 10^{14}$ GeV [42]. This is below the Planck scale by

approximately four orders of magnitude, a fact that will prove critical when the asymptotic-safety constraints are applied [47].

5.2 Obstacles to UV Completion of the Abelian Higgs Model

The Abelian Higgs model in the presence of gravity presents two principal obstacles to asymptotic-safety completion [47]. The first concerns the fixed-point structure of the coupled gauge-scalar system. In the absence of gravity, the $U(1)$ gauge coupling has a Landau pole: it grows without bound at high energies, indicating a breakdown of the theory in the UV. Gravitational antiscreening can in principle counteract this tendency and generate an interacting fixed point for the gauge coupling, but the detailed analysis shows that the interplay between gravitational fluctuations and the gauge-scalar sector does not generically produce a fixed point with a perturbatively stable scalar potential.

The second obstacle is a scale mismatch. Even in the presence of a free (Gaussian) fixed point for the gauge sector, the symmetry-breaking VEV is driven by the RG flow to the Planck scale:

$$v \sim M_{\rm Pl} \approx 10^{19}\,\text{GeV}. \tag{19}$$

This is in severe conflict with the observational bound of $v \lesssim 4.6 \times 10^{14}$ GeV from NANOGrav [42]. The minimal Abelian Higgs model, when embedded in asymptotic safety, produces cosmic strings whose tension is vastly too large to be consistent with PTA data.

5.3 The Fermion Requirement: The Forty-Two Dirac Fermion Barrier

To force the symmetry-breaking scale below the Planck mass, one may extend the model by coupling N_f Dirac fermions charged under the $U(1)$ gauge symmetry. Fermions contribute to the running of the gauge coupling with a sign opposite to that of gauge bosons: they screen the coupling and can reduce the fixed-point value of the gauge coupling g_*^2 relative to the Yukawa coupling y_*^2, thereby shifting the locus of spontaneous symmetry breaking to lower energies [47].

The analysis of the stability conditions reveals two distinct thresholds. For the scalar potential to admit any real solution for the mass parameter m_*^2 below the Planck scale, one needs at least $N_f \geq 22$ Dirac fermions. However, full stability across the cosmologically relevant range of the dimensionless cosmological constant requires

$$N_f \geq 42. \tag{20}$$

This result converts a qualitative incompatibility into a precise particle-content requirement: the simplest field-theoretic model of cosmic strings demands at least forty-two additional charged Dirac fermions beyond the Standard Model merely to satisfy the UV consistency conditions imposed by asymptotic safety. From a model-building perspective, this is an extremely high price for a simple infrared phenomenon—what one might aptly call a complexity tax [47].

5.4 Alignment with NANOGrav

The theoretical obstacles to accommodating cosmic strings in asymptotic safety find striking observational support in the NANOGrav 15-year dataset [42]. This dataset provides strong evidence for a stochastic gravitational-wave background (SGWB) via the detection of quadrupolar Hellings-Downs angular correlations between pulsars. The spectral properties of the observed signal, however, are inconsistent with the characteristic spectrum expected from a network of stable field-theoretic cosmic strings. The data prefer instead a population of supermassive black-hole binary systems in the final stages of inspiral.

This observational preference aligns with the theoretical expectation from asymptotic safety [47]: the minimal models that would produce a detectable cosmic-string background are precisely those that are theoretically inconsistent without a dramatic extension of the matter sector. The convergence of top-down theoretical constraints and bottom-up observational evidence strengthens the case that simple field-theoretic cosmic strings are not a generic prediction of UV-complete theories, but a feature of fine-tuned or theoretically problematic constructions.

6 The Small Landscape: UV Consistency as a Structural Filter

6.1 Theory Space and the UV Critical Surface

The results described in the preceding sections instantiate a general principle: the requirement of asymptotic safety dramatically reduces the dimensionality of the viable parameter space [13, 15]. The full theory space $\mathcal{T}$ is infinite-dimensional. The asymptotically safe fixed point defines a UV-critical surface: the set of all RG trajectories that pass through or terminate at the fixed point as $k \to \infty$. Points not on this surface cannot be connected to the fixed point and therefore do not correspond to UV-complete theories.

The dimensionality of the UV-critical surface equals the number of relevant directions at the fixed point [17, 16]. In the Einstein-Hilbert truncation this is two,

corresponding to the Newton coupling and the cosmological constant. Including matter fields typically adds a small number of additional relevant directions. The key point is that this dimensionality is finite, while $\mathcal{T}$ is infinite-dimensional. The UV-critical surface is therefore a measure-zero subset of theory space. The failure of kinetic braiding and simple cosmic strings is not surprising; it is the generic expectation, with the remarkable thing being if an arbitrarily chosen low-energy model happened to lie on the UV-critical surface.

6.2 Predictive Power and Irrelevant Couplings

The mechanism through which asymptotic safety constrains phenomenology is the irrelevance of couplings at the UV fixed point. An irrelevant coupling is not physically unimportant: it is predicted. Its infrared value is determined by the RG flow from the fixed point, not by a free parameter that must be measured [2, 10]. When the predicted infrared value conflicts with what is observationally required, the model is excluded.

This mechanism has been spectacularly successful in the gravitational sector. The quartic Higgs self-coupling is irrelevant at the gravitational NGFP, meaning that its value at the electroweak scale is predicted by the RG flow from the Planck scale. This prediction yields $M_H \approx 126\,\text{GeV}$ [15, 63], in remarkable agreement with the measured value of $125.25\,\text{GeV}$. The braiding coupling h and the cosmologically required value of g fail an analogous test in the opposite direction: the predicted values are incompatible with observation. The same framework has been applied to Standard Model couplings beyond the Higgs sector, including the Yukawa coupling of the top quark [48], the gauge coupling of hypercharge [67], and the quark and lepton mixing angles [55].

6.3 A Unified Constraint Mechanism

The two phenomenological examples examined in this essay—shift-symmetric Horndeski dark energy and Abelian Higgs cosmic strings—illustrate a common structural mechanism by which asymptotic safety constrains infrared physics. In both cases the decisive ingredient is the classification of couplings as relevant or irrelevant at the ultraviolet fixed point [17, 18].

For the Horndeski sector, the braiding coupling h is symmetry-protected and irrelevant, forcing its infrared value to vanish [65]. For the Abelian Higgs model, the scalar potential parameters are driven toward Planck-scale values unless additional fermionic matter is introduced to modify the RG flow [47]. These examples demonstrate that the ultraviolet consistency requirement acts as a structural filter on theory space [60]. Phenomenological models that appear viable at low

energies may fail when embedded in a UV-complete framework. The asymptotic-safety paradigm therefore reduces the apparent freedom of model building and narrows the landscape of consistent theories.

6.4 Comparison with the Swampland Program

The asymptotic-safety constraints described in this essay bear a structural resemblance to the swampland conjectures of string theory [49]. Both programs impose top-down consistency conditions on effective field theories, restricting the space of viable low-energy models. The swampland program typically invokes the absence of global symmetries, the absence of stable non-supersymmetric vacua, or the Trans-Planckian Censorship Conjecture to argue that many apparently consistent EFTs cannot be embedded in a UV-complete framework [49].

The asymptotic-safety filter operates differently but achieves a similar result through a more concrete, computational mechanism. Rather than invoking general consistency arguments about quantum gravity, it works through explicit fixed-point analysis of the coupled gravity-matter FRG flow. Its predictions are therefore more specific and, in principle, more falsifiable. The braiding coupling is not merely argued to be small; it is computed to be exactly zero. The fermion count required for cosmic string viability is not merely bounded below; it is fixed to a specific integer threshold. This concreteness is one of the most attractive features of the asymptotic-safety approach to the swampland problem [49, 18].

7 Methodology: The FRG Toolchain for Gravity-Matter Systems

7.1 Truncation Strategy

The practical implementation of the FRG for gravity-matter systems begins with the choice of truncation [10, 11, 12]. The general principle is that operators with lower canonical dimension dominate the flow near the Gaussian or near-Gaussian fixed point, so truncations including all operators up to a given canonical dimension capture the most important physics. In practice, the Einstein-Hilbert sector with Newton coupling and cosmological constant is the universal starting point, supplemented by matter kinetic terms and the specific interaction operators relevant to the model under study.

For the Horndeski analysis, the truncation described in Section 4.2 includes the full set of shift-symmetric operators up to the cubic level, with the quartic

sector included as a robustness check through the coupling ζ. The decomposition of metric fluctuations into transverse-traceless, trace, and longitudinal modes uses the York decomposition (Appendix B), which separates the spin-2 transverse-traceless modes carrying the primary gravitational degrees of freedom from the spin-0 trace mode that couples to the scalar field through the $\zeta\chi R$ operator.

7.2 The $P^{-1}F$ Expansion

The evaluation of the right-hand side of the Wetterich equation requires computing the trace of a matrix involving the full inverse propagator $\Gamma_k^{(2)} + R_k$. This is accomplished through the $P^{-1}F$ expansion [11, 12]:

$$(\Gamma_k^{(2)} + R_k)^{-1} = (P + F)^{-1} = P^{-1} \sum_{n=0}^{\infty} (-FP^{-1})^n, \tag{21}$$

where $P = \Gamma_k^{(2)} + R_k|_{\Phi=0}$ is the propagator evaluated at a background field configuration and $F = \Gamma_k^{(2)} - \Gamma_k^{(2)}|_{\Phi=0}$ is the fluctuation-field-dependent part. The trace of each term in this expansion can be evaluated using heat-kernel methods detailed in Appendix C, or by direct momentum integration with the Litim regulator. The extraction of beta functions by projection onto specific field monomials is described in Appendix D. The approximation of neglecting gravitational and scalar anomalous dimensions, justified near the shifted Gaussian fixed point where canonical scaling dominates, simplifies the computation while preserving the qualitative structure of the results [11, 12].

7.3 Truncation Robustness and Regulator Dependence

All practical implementations of the FRG require truncation of the infinite-dimensional theory space. The reliability of results must therefore be assessed through robustness checks [10, 12].

Regulator dependence can be examined by repeating the calculation with different infrared cutoff functions R_k. Although numerical values of fixed-point couplings may vary slightly, the existence of the fixed point and the sign of the critical exponents should remain stable [10]. Gauge dependence is controlled through the background-field method and the choice of gauge-fixing parameters. The Landau-DeWitt gauge used in this work simplifies the propagator structure while preserving background diffeomorphism invariance [14, 16].

Truncation stability can be tested by enlarging the truncation to include higher-order operators. Existing studies of asymptotically safe gravity-matter systems indicate that the existence of the gravitational fixed point and the irrelevance of

higher-derivative scalar operators are stable features across a wide class of truncations [18, 45]. The symmetry-protection argument for $h_* = 0$ is particularly robust: it does not depend on the specific numerical values of coefficients in the truncation, but only on the discrete $\mathbb{Z}_2$ symmetry that the cubic operator breaks. Any truncation preserving this symmetry will find $h_* = 0$ [65].

8 Observational Connections and the Narrowing Landscape

8.1 NANOGrav and the Gravitational Wave Background

The 15-year dataset of the NANOGrav pulsar timing array provides the most significant recent observational input for the theoretical constraints described in this essay [42]. The detection of a stochastic gravitational-wave background via Hellings-Downs angular correlations confirms that PTAs are now sensitive instruments for cosmological physics. The spectral index of the observed signal is critical: a network of field-theoretic cosmic strings in the scaling regime produces a background with a characteristic spectral shape determined by the string tension and the loop distribution function. The NANOGrav data prefer a steeper spectrum more consistent with a population of supermassive black-hole binaries in the nanohertz frequency band.

This observational preference aligns with the theoretical expectation from asymptotic safety [47]. The minimal Abelian Higgs model produces strings with tension at the Planck scale, far too large to be consistent with PTA constraints. The convergence of these two independent lines of evidence—one top-down from quantum gravity and one bottom-up from observation—represents a powerful joint constraint on the cosmic string hypothesis. It exemplifies the connection noted by Eichhorn, Lino dos Santos, and Miqueleto [47]: quantum gravity constraints on the particle-physics models that can give rise to a stochastic GW background become directly testable through PTA observations.

8.2 Gravitational Wave Astronomy as a Probe of Quantum Gravity

The connection between asymptotic safety and the gravitational wave background represents a methodological advance of significant importance. It demonstrates that constraints derived from UV consistency at the Planck scale can be translated into predictions for astrophysical observables in the nanohertz frequency

band. The bridge is the RG flow itself: the UV fixed-point structure constrains the symmetry-breaking scale of the Abelian Higgs model, which in turn determines the string tension, which determines the amplitude and spectral shape of the gravitational wave background [47].

This chain of reasoning suggests that gravitational wave observatories, from PTAs to LIGO [43] to future space-based detectors, are not merely instruments for studying astrophysical compact objects but probes of the fundamental structure of quantum gravity. A precise measurement of the stochastic background spectrum can, in principle, discriminate between field-theoretic string models embedded in different UV completions and thereby test the predictions of asymptotic safety at cosmological scales [60, 44].

8.3 Black Holes, Regular Geometries, and the Asymptotically Safe Landscape

Beyond cosmic strings and dark energy, the asymptotic-safety program has made recent progress on other observational frontiers. The possibility of regular black holes—spacetimes without a central singularity—has been studied within asymptotically safe gravity through RG-improved metrics [56, 68]. Such regular geometries can in principle be distinguished from classical black holes through deviations in the shadow morphology visible to the Event Horizon Telescope [66]. The positivity bounds derived from unitarity and causality have also been applied within asymptotically safe gravity to constrain the parameter space of the effective theory [59], and constraints on ultralight dark matter have been derived from the gravitational fixed-point structure [57].

9 Beyond the Current Paradigm: Open Questions and Future Directions

9.1 Non-Polynomial Dynamics and the Deep IR

The most significant theoretical limitation of current FRG analyses is the reliance on polynomial truncations for the scalar sector [10, 12]. The full effective potential $V(\phi)$ is an infinite-dimensional function, and truncating it to a polynomial in ϕ^2 captures only the local behavior near the expansion point. Non-polynomial fixed-point potentials, of the type found in scalar field theories using the local potential approximation of the FRG, can exhibit qualitatively different behavior, including the possibility of spontaneous symmetry breaking at scales well below

the truncation scale [20, 22].

For the Horndeski dark energy problem, a non-polynomial treatment of the kinetic sector would amount to considering the general function $G_2(\chi)$ in the Horndeski Lagrangian rather than its truncation to a quadratic polynomial. Whether such a treatment can reconcile asymptotic safety with dynamical dark energy is an open and important question. The technical challenge is substantial: the flow equation for a non-polynomial function is a functional partial differential equation in field space, and its analysis requires either numerical methods or specific ansätze for the functional form of the potential.

9.2 Lorentzian Signature and Causal Structure

All existing FRG analyses of asymptotically safe gravity are performed in Euclidean signature, where the metric is positive-definite and the path integral is well-defined as a statistical sum. The connection to physical Lorentzian spacetimes requires an analytic continuation that is well-understood for static backgrounds but becomes problematic for dynamical cosmological spacetimes where the causal structure is essential. Recent work on spin-foam models and tensor field theories has suggested possible approaches to Lorentzian FRG formulations [50, 61], but a complete implementation for the full gravity-matter system remains outstanding. This is particularly relevant for cosmological applications, where the predictions of the UV-fixed-point analysis must be connected to the real-time evolution of the universe.

9.3 Area-Metric Gravity and Extended Geometries

A further direction that may extend the reach of asymptotic safety is the consideration of gravitational theories based on more general geometric structures than the metric. Area-metric gravity, in which the fundamental variable is an area metric $G_{\mu\nu\rho\sigma}$ rather than the standard metric $g_{\mu\nu}$, provides more degrees of freedom and may admit richer fixed-point structures [54]. Scalar-Gauss-Bonnet gravity with higher-order couplings [62] extends the Horndeski sector in directions not covered by the shift-symmetric truncation analyzed here. These extensions may provide the additional structure needed to reconcile asymptotic safety with specific phenomenological requirements, or they may impose even more stringent constraints, advancing the understanding of the asymptotically safe landscape in either case.

9.4 From Proton Decay to Neutrino Masses: Matter-Gravity Unification

The scope of asymptotic safety predictions extends well beyond the models discussed in detail here. Recent work has established that the framework makes predictions for proton stability [46], neutrino mass generation [51], and the possible existence of axion-like particles [67]. The question of CPT symmetry restoration has been addressed within the framework [52], and higher-curvature operators in causal set quantum gravity have been studied as a complementary UV-completion strategy [64]. The initial distribution of Standard Model couplings at the Planck scale has been constrained by the structure of asymptotically safe trajectories [53].

The breadth of these applications suggests that asymptotic safety is evolving from a specific hypothesis about the UV behavior of quantum gravity into a comprehensive framework for connecting Planck-scale physics to phenomenology across all sectors of the Standard Model and its extensions [60, 45]. Whether this program will ultimately yield a unique theory or a family of consistent completions depends on questions about the global structure of theory space—the existence and classification of universality classes of gravity-matter systems [61]—that are not yet settled.

10 Diagrammatic Structure of Graviton-Matter Loop Contributions

10.1 One-Loop Contributions to Matter Beta Functions

The beta functions for matter couplings in an asymptotically safe gravity-matter system receive contributions from two distinct sources: self-interactions of the matter fields, which reproduce the flat-spacetime running, and gravitational fluctuations mediated by virtual graviton exchange. Understanding the diagrammatic structure of these contributions clarifies why gravity can promote irrelevant couplings toward relevance and why the symmetry-protection argument for the braiding coupling is particularly robust [14, 10, 12].

Within the one-loop nonperturbative structure of the Wetterich equation, the trace on the right-hand side of equation (1) generates, after expanding in the $P^{-1}F$ framework, a set of contributions that can be organized diagrammatically. Consider a generic matter coupling g_i appearing as a vertex in the action. The gravitational contribution to β_{g_i} arises from diagrams in which two matter field lines are connected by a graviton propagator. Schematically, one encounters two classes of diagram.

The first class involves a single graviton loop closing between two matter vertices. This contribution is proportional to $g_i \cdot G_N/(k^2 + m^2_{\text{grav}})$, where G_N is the Newton coupling and $m^2_{\text{grav}} = 2\Lambda k^2$ is the effective graviton mass generated by the cosmological constant [16, 15]. This term modifies the anomalous dimension of the matter field and shifts the canonical scaling of g_i. For couplings with negative canonical dimension $d_i < 0$, gravity can in principle compensate the power-counting suppression and make the coupling marginal or relevant.

The second class involves the seagull diagrams generated by the minimal coupling of matter to gravity through the covariant derivative and the metric determinant $\sqrt{g}$. These diagrams produce contact interactions between the matter sector and the graviton propagator without introducing additional matter vertices. They contribute to the gravitational shift of fixed-point values but do not alter the symmetry-based selection rules on which operators can be generated [18].

10.2 The Gravitational Shift of Scalar Couplings

For the scalar sector of the Horndeski truncation, the gravitational contributions to the beta functions of h and g can be isolated by writing

$$\beta_h = \beta_h^{\text{matter}} + \beta_h^{\text{grav}}, \tag{22}$$

$$\beta_g = \beta_g^{\text{matter}} + \beta_g^{\text{grav}}. \tag{23}$$

The matter contributions β_h^{matter} and β_g^{matter} reproduce the flat-spacetime running of these couplings in the absence of gravity. In the shift-symmetric sector, $\beta_h^{\text{matter}} \propto h$ by the symmetry argument of Section 3.4, and β_g^{matter} receives contributions from the χ^2 self-interaction and from the mixing of h and g through intermediate field configurations.

The gravitational contribution β_h^{grav} also inherits the $\mathbb{Z}_2$ symmetry constraint: because gravity couples to the scalar field only through $\chi = -\frac{1}{2}\nabla_\mu\phi\nabla^\mu\phi$, which is $\mathbb{Z}_2$-even, any graviton-mediated loop diagram connecting to the h-vertex must produce a factor proportional to h. The gravitational sector therefore cannot break the $\mathbb{Z}_2$ symmetry protection of the braiding coupling [65, 18]. This is the diagrammatic expression of the symmetry argument: gravity mediates interactions, but it cannot generate new quantum numbers that the matter sector does not already carry.

10.3 Antiscreening from Graviton Exchange

The gravitational contribution to β_g is qualitatively different: because g is $\mathbb{Z}_2$-even, the graviton loop can generate a non-zero contribution to β_g that does not vanish

at $g = 0$. This contribution takes the form

$$\beta_g^{\text{grav}} \supset \frac{c_g\, G}{(1-2\Lambda)^p}, \tag{24}$$

where c_g is a positive numerical coefficient determined by the graviton spin content and p depends on the regulator scheme [16, 65]. This antiscreening contribution shifts the fixed-point value of g away from zero—the Gaussian fixed point—to a non-zero value $g_* \neq 0$. The result is the shifted Gaussian fixed point: a fixed point in which $h_* = 0$ is maintained by symmetry while g_* is shifted to a positive value by gravitational antiscreening.

The magnitude of this shift is controlled by the gravitational coupling strength G at the fixed point. Near the weak-gravity bound, where $G \to G_{\text{crit}}$, the shift can become large enough to change the sign of the critical exponent θ_g, potentially promoting g from an irrelevant to a marginal or relevant coupling. However, as shown by the detailed analysis of [65], the resulting IR value of g remains many orders of magnitude below the phenomenologically required value even in this near-critical regime.

11 Operator Dimension Counting and the Truncation Landscape

11.1 Canonical Dimensions and Power Counting

A systematic understanding of which operators can appear at a given order in the FRG truncation requires careful power counting [2, 3, 7]. In four spacetime dimensions, an operator $\mathcal{O}$ with canonical mass dimension $[\mathcal{O}] = d$ has a coupling constant of canonical dimension $4 - d$ (from the requirement that the action be dimensionless). Operators with $d > 4$ are power-counting irrelevant and their couplings have negative mass dimension; operators with $d < 4$ are relevant; operators with $d = 4$ are marginal.

Table 1 lists the canonical dimensions and classical scaling properties of the operators appearing in the Horndeski truncation studied in this work, together with an indication of whether asymptotic safety predicts each coupling to be relevant or irrelevant at the SGFP.

11.2 Beyond the Truncation: Convergence Arguments

The operators listed in Table 1 constitute the minimal truncation required to capture the phenomenology of kinetic braiding dark energy. The question of whether

Operator	Symbol	Canonical dim. of coupling	Classical scaling	AS verdic
Cosmological constant	Λ	$+2$	Relevant	Free param
Einstein-Hilbert	R	0	Marginal	Free param
Scalar kinetic	χ	0	Marginal	Free param
Non-minimal coupling	$\zeta\chi R$	0	Marginal	Shifted, irre
Cubic braiding	$h\chi\Box\phi$	-3	Irrelevant	$h_* = 0$ (sym
Quartic interaction	$g\chi^2$	-4	Irrelevant	$g_* \neq 0$ (shifte
Higher kinetic	χ^3	-8	Strongly irrel.	≈ 0

Table 1: Canonical dimensions and asymptotic-safety predictions for operators in the shift-symmetric Horndeski truncation. The column "AS verdict" indicates whether the coupling is a free parameter (relevant), symmetry-protected to zero, or shifted to a small non-zero value by gravitational antiscreening. Strongly irrelevant operators are suppressed by additional powers of $(k/M_{\rm Pl})$ and are negligible in the near-Planck regime.

extending this truncation to include higher-order operators would qualitatively alter the conclusions can be addressed through several arguments [10, 12, 11].

First, the symmetry-protection argument for $h_* = 0$ is exact within any truncation that preserves the $\mathbb{Z}_2$ symmetry. Adding higher-order $\mathbb{Z}_2$-odd operators to the truncation would produce additional symmetry-protected couplings, all vanishing at the fixed point, but would not change the fate of h itself. Second, the magnitude mismatch for g—fifty-seven orders of magnitude between the AS prediction and the cosmologically required value—is so large that it is not plausibly an artifact of the truncation. Even if the non-polynomial treatment of $G_2(\chi)$ shifted g_* by several orders of magnitude, it would not bridge a gap of 10^{57} [65].

Third, the convergence of FRG truncations in the scalar sector has been studied extensively in the context of $O(N)$ models and related theories [10, 22, 20]. In these cases the local potential approximation, which retains the full non-polynomial potential $V(\phi)$ while truncating the derivative sector, provides excellent quantitative agreement with exact results. Applying this lesson to the Horndeski system suggests that extending the kinetic truncation to the full function $G_2(\chi)$ is the most important next step, while the qualitative predictions of the polynomial analysis remain reliable at the level of order-of-magnitude estimates.

12 The RG Phase Diagram: From the Planck Scale to the Cosmic Horizon

12.1 Structure of the Gravitational Phase Diagram

The renormalization group flow of asymptotically safe gravity-matter systems can be visualized as a phase diagram in the space of dimensionless couplings (G, Λ), supplemented by the matter coupling directions [15, 16, 17]. The phase diagram organizes all RG trajectories and makes transparent which initial conditions in the UV lead to viable low-energy physics.

In the Einstein-Hilbert sector alone, the phase diagram in the (G, Λ) plane exhibits several qualitatively distinct regions. The non-Gaussian fixed point (NGFP) sits at a specific location (G_*, Λ_*) with $G_* > 0$ and $\Lambda_* > 0$, determined by the condition $\beta_G = \beta_\Lambda = 0$. The UV-critical surface is a two-dimensional manifold passing through this fixed point, consisting of all RG trajectories that approach it as $k \to \infty$. Trajectories that do not lie on this surface diverge from the NGFP and either reach a singularity at finite scale or flow to a Gaussian fixed point at $(G, \Lambda) = (0, 0)$.

Among the UV-critical trajectories, one must further identify those that produce physically acceptable IR behavior: a positive effective Newton constant, a small positive cosmological constant consistent with observation, and matter sector couplings that match the known parameters of the Standard Model. This additional requirement selects a subset of the UV-critical surface, reducing the viable parameter space to what has been called the "trajectory of the universe" in the asymptotic-safety literature [15, 19].

12.2 Matter Coupling Trajectories

When matter fields are included, the phase diagram extends into a higher-dimensional space. The gravitational coupling directions remain (G, Λ), while the matter sector introduces additional dimensions for each independent coupling [17, 18]. The structure of the RG flow in this extended space determines the fate of each matter coupling.

For the Horndeski system, the relevant part of the phase diagram can be described in terms of the three-dimensional space (G, h, g) at fixed $\Lambda = 0$ for simplicity. The SGFP is located at $(G_*, 0, g_*)$. The UV-critical surface passes through this point with a two-dimensional tangent plane spanned by the directions corresponding to G and the linear combination of h and g that is relevant at the fixed point. Since both h and g are irrelevant at the SGFP, the UV-critical surface is actually two-dimensional (the G and Λ directions), and trajectories that start with

$h \neq 0$ in the UV are driven toward $h = 0$ as $k \to \infty$ [65].

Flowing from the UV fixed point toward the IR, trajectories initially move along the two-dimensional UV-critical surface in the (G, Λ) plane. As the scale k decreases through the Planck scale $M_{\rm Pl}$, the dimensionless couplings begin to deviate from their fixed-point values. The Newton coupling $G = G_N k^2$ decreases as k^2, reproducing Newton's law at macroscopic scales. The cosmological constant $\Lambda = \bar{\Lambda}/k^2$ grows at low scales, which in the dimensionful theory corresponds to a small but non-zero value of $\bar{\Lambda}$ consistent with observation [19, 40].

12.3 The Separation of Scales and the Dark Energy Problem

The most striking feature of the RG phase diagram for gravity-matter systems is the enormous separation of scales between the UV fixed point at $k \sim M_{\rm Pl}$ and the infrared regime relevant for dark energy at $k \sim H_0$. This separation spans approximately sixty orders of magnitude in energy:

$$\frac{M_{\rm Pl}}{H_0} \approx \frac{10^{19}\,\text{GeV}}{10^{-42}\,\text{GeV}} = 10^{61}. \tag{25}$$

During this enormous RG evolution, irrelevant couplings are suppressed by powers of $(k/M_{\rm Pl})^{|\theta_I|}$. For a coupling with critical exponent $|\theta_I| \sim 1$, the suppression factor at the scale H_0 is $(H_0/M_{\rm Pl}) \approx 10^{-61}$. This is the origin of the fifty-seven-order-of-magnitude mismatch for g: the canonical dimension of g is -4, so the suppression at the cosmological scale is $(H_0/M_{\rm Pl})^4 \sim 10^{-244}$ from canonical scaling alone, partially offset by the gravitational shift of the fixed-point value [65].

This separation of scales is not a problem to be solved but a structural feature of the theory that explains why the cosmological constant problem is so severe from the EFT perspective: any mass scale in the theory is naturally of order $M_{\rm Pl}$, and bringing it down to H_0 requires either enormous fine-tuning or a symmetry mechanism. The asymptotic-safety prediction is that no such mechanism exists within the shift-symmetric Horndeski framework. This is not a failure of the paradigm; it is the paradigm successfully ruling out a class of theories that could not have worked [49, 60].

12.4 Entropy, Information Loss, and Geometric Flows

The structure of the RG phase diagram carries a deeper geometric interpretation that connects the asymptotic-safety program to foundational work in geometric analysis and nonequilibrium thermodynamics [23, 24, 26]. The renormalization group flow is, in a precise sense, an information-loss process: as the scale k is

lowered, short-distance fluctuations are integrated out and the effective description of the system becomes coarser. The information content of the UV theory is progressively compressed into a smaller number of relevant parameters [25, 20].

This observation connects the FRG program to the mathematical theory of geometric flows initiated by Hamilton's work on Ricci flow [69] and subsequently developed by Perelman's proof of the geometrization conjecture [70]. In Ricci flow the metric of a Riemannian manifold evolves according to

$$\partial_t g_{\mu\nu} = -2R_{\mu\nu}, \tag{26}$$

driving the geometry toward spaces of constant curvature—the fixed points of the flow. The analogy with the RG is precise: just as Ricci flow eliminates geometric irregularities to produce canonical geometries, the RG flow eliminates short-scale fluctuations to produce scale-invariant effective theories. The fixed points of Ricci flow (Einstein manifolds) correspond to fixed points of the RG (scale-invariant quantum field theories) [27, 28].

Perelman's key insight was to introduce the $\mathcal{F}$-functional

$$\mathcal{F}(g, f) = \int_M (R + |\nabla f|^2)\, e^{-f}\, dV, \tag{27}$$

which is monotonically non-decreasing along the Ricci flow and serves as a Lyapunov functional ensuring convergence [70]. The analogous object in the RG context is Zamolodchikov's c-function in two dimensions, or more generally the proposed a-function in four dimensions, which decreases monotonically along RG flows [6, 5]. Whether a monotone Lyapunov functional exists for the full gravity-matter RG flow of asymptotic safety remains an open question, but the geometric analogy suggests that such a functional, if it exists, would take the form of an entropy-like quantity defined on theory space [73, 71].

The nonequilibrium thermodynamic perspective, pioneered by Onsager [71] and later systematized by Prigogine [72] and developed into an information-theoretic framework by Jaynes [73], interprets irreversible processes as entropy-generating flows toward states of maximum entropy. The RG flow, viewed as an irreversible coarse-graining process, fits naturally into this framework: information is lost monotonically as the resolution scale decreases, and fixed points are states of maximum entropy compatible with the symmetry constraints of the theory [25, 24]. This interpretation does not alter the technical content of the FRG analysis but suggests a broader conceptual architecture in which asymptotic safety, geometric flow theory, and nonequilibrium thermodynamics are unified aspects of a single information-geometric framework [26, 23].

13 Consistency Checks and Comparison with Other Quantum Gravity Constraints

The asymptotic-safety constraints discussed in this work are part of a broader effort to identify general principles that restrict the space of low-energy effective field theories compatible with quantum gravity. Several independent approaches have been developed in recent years, including the string-theoretic swampland program [49], effective field theory positivity bounds [59], and semiclassical consistency conditions. Although these frameworks arise from different theoretical starting points, they share a common goal: to distinguish effective theories that can arise from a consistent ultraviolet completion from those that cannot. In this section we briefly compare the constraints derived from asymptotic safety with those obtained from these alternative approaches.

13.1 Relation to the Swampland Program

The swampland program [49], originally proposed in the context of string theory, seeks to characterize the set of effective field theories that cannot be embedded into a consistent theory of quantum gravity. Theories that fail these criteria are said to lie in the swampland rather than in the landscape of viable quantum gravity vacua. Many swampland conjectures place constraints on scalar-field potentials, derivative interactions, and the range of validity of effective field theories. For example, the de Sitter conjecture suggests that scalar potentials compatible with quantum gravity must satisfy bounds of the form $|\nabla V| \geq c\,V$, where c is an $\mathcal{O}(1)$ constant in Planck units.

Although the asymptotic-safety framework does not assume string theory or extra dimensions, it produces conceptually similar restrictions through an entirely different mechanism [17, 60]. Couplings that are irrelevant at the ultraviolet fixed point are not free parameters but predictions of the theory. If the predicted infrared value of such a coupling conflicts with the requirements of a phenomenological model, the model is excluded. In this sense, asymptotic safety provides an explicit dynamical mechanism by which certain low-energy theories are removed from the space of UV-complete models, complementing the more axiomatic constraints of the swampland program [49].

13.2 Positivity Bounds from Effective Field Theory

Another independent set of constraints arises from analyticity and unitarity of scattering amplitudes [59]. In local, Lorentz-invariant quantum field theories, dis-

persion relations imply positivity conditions on the coefficients of higher-derivative operators in the low-energy effective action. For scalar field theories the coefficient of the operator $(\partial_\mu\phi\,\partial^\mu\phi)^2$ must be non-negative to ensure that forward scattering amplitudes obey the optical theorem. Recent work has extended these bounds to gravity-matter systems within asymptotically safe gravity [59]. These analyses show that positivity bounds can be compatible with the existence of a non-Gaussian fixed point but may restrict the allowed range of higher-derivative couplings along RG trajectories emerging from the fixed point. In this sense, positivity bounds provide complementary infrared consistency checks on the RG flows predicted by asymptotic safety.

13.3 Semiclassical Gravity Constraints

A third class of consistency conditions arises from semiclassical gravity. In semiclassical treatments, quantum matter fields propagate on a classical gravitational background, and their stress-energy tensor acts as a source in the Einstein equations. Semiclassical consistency requires that expectation values of the stress-energy tensor satisfy certain energy conditions and stability criteria. The asymptotic-safety framework automatically incorporates many of these constraints because the RG flow controls the sign and magnitude of kinetic and interaction terms in the effective action [16, 15]. Operators that would lead to pathological behavior—such as wrong-sign kinetic terms or unstable potentials—typically correspond to directions in theory space that do not lie on the UV-critical surface. The recent work on regular black holes and gravastars [56, 68] represents a concrete application of this principle: the RG-improved geometry removes the central singularity precisely because the running Newton coupling suppresses gravitational strength at short distances, maintaining semiclassical consistency across the interior.

13.4 Convergence of Independent Constraints

Taken together, these different approaches suggest a coherent picture. Independent theoretical principles—renormalization group fixed points, string-theoretic consistency conditions, analyticity of scattering amplitudes, and semiclassical stability—tend to restrict effective field theories in qualitatively similar ways [60, 18]. The asymptotic-safety paradigm provides a concrete realization of this idea by explicitly constructing renormalization group flows that connect ultraviolet fixed points to infrared physics. The examples studied in this essay illustrate how this connection translates Planck-scale consistency conditions into observable constraints on cosmological models, and the convergence of these independent lines of argument strengthens the case that the asymptotically safe landscape is genuinely small.

14 Renormalization Group Flow as Coarse-Graining of Physics

The renormalization group can be understood intuitively as a systematic procedure for coarse-graining physical systems [2, 21, 20]. At any given scale k, the effective action Γ_k describes the dynamics of fluctuations with momenta smaller than k. Fluctuations with higher momentum have already been integrated out. Lowering the scale k corresponds to gradually averaging over shorter-distance physics. In this sense, the renormalization group provides a continuous interpolation between microscopic and macroscopic descriptions of nature.

This coarse-graining picture was originally developed in the context of statistical physics, where it explains the universality of critical phenomena [1, 21, 22, 6]. Systems with very different microscopic structures can exhibit identical large-scale behavior because the renormalization group flow drives their couplings toward the same fixed point. The universality classes of critical phenomena—characterized by the fixed-point structure and critical exponents—are among the most striking examples of emergent simplicity from underlying complexity.

The asymptotic-safety scenario applies this logic to quantum gravity [13, 14]. Instead of describing the infrared behavior of condensed matter systems, the renormalization group flow is followed toward extremely high energies. If the flow approaches a fixed point as $k \to \infty$, the theory remains well-defined at arbitrarily short distances. Ultraviolet completion is not achieved by introducing new degrees of freedom but by the self-organizing structure of the renormalization group flow itself, which is precisely the appeal of the program from a minimalist perspective [17, 16].

15 Why Fixed Points Generate Predictivity

The predictive power of an asymptotically safe theory arises directly from the structure of the ultraviolet fixed point. Near a fixed point, many couplings are attracted toward specific values as the energy scale increases [2, 10]. These couplings are called irrelevant because their ultraviolet behavior is not sensitive to their initial values: any nearby trajectory is drawn toward the fixed-point value. From the perspective of low-energy physics, this means that the infrared values of irrelevant couplings are determined by the RG trajectory emerging from the fixed point rather than by independent free parameters.

A useful analogy comes from dynamical systems theory. A stable fixed point in a phase portrait attracts trajectories from a neighborhood of initial conditions.

Even if the initial values of the couplings differ, they are drawn toward the same fixed-point value as the energy scale increases. In asymptotically safe quantum gravity, the UV fixed point plays precisely this role: only a small number of directions in theory space allow trajectories to leave the fixed point toward lower energies, corresponding to the relevant couplings that must be measured experimentally. All other couplings are predicted [17, 15]. The irrelevance of the Horndeski braiding coupling h is the specific realization of this mechanism in the dark energy sector [65].

16 Quantum Gravity as a Structural Constraint on Cosmological Models

The examples discussed in this essay illustrate a broader principle: cosmological model building cannot be separated from the ultraviolet structure of fundamental physics. In traditional effective field theory approaches, cosmological models are constructed by writing down the most general Lagrangian consistent with symmetries at the relevant energy scale. Parameters are adjusted to reproduce observational data. This strategy has proven extremely successful, but it implicitly assumes that the effective theory can be extended to arbitrarily high energies without encountering inconsistencies [3, 5].

Quantum gravity challenges this assumption. The renormalization group structure of asymptotically safe gravity implies that not every low-energy effective theory can be connected to a consistent ultraviolet completion [13, 14, 17]. Only those theories lying on RG trajectories that originate from the ultraviolet fixed point are admissible. From this perspective, ultraviolet consistency acts as a structural constraint on cosmological model building: instead of beginning with the most general Lagrangian allowed by symmetry, one must restrict attention to theories whose parameters can arise from renormalization group flows compatible with the fixed-point structure of quantum gravity.

The two case studies analyzed in this work illustrate how such constraints operate in practice. In the shift-symmetric Horndeski sector, the renormalization group flow forces the braiding coupling to vanish because it is symmetry protected at the ultraviolet fixed point [65]. In the cosmic string scenario, the Abelian Higgs model admits spontaneous symmetry breaking only at the Planck scale when embedded in the asymptotically safe framework, and lower symmetry-breaking scales require the introduction of a large number of additional fermionic degrees of freedom [47]. Both examples reveal the same underlying mechanism: the ultraviolet fixed point determines which directions in theory space are phys-

ically accessible, and couplings that lie outside this set cannot be realized by any RG trajectory connected to the fixed point.

17 Implications for the Strategy of Cosmological Model Building

The analysis presented in this work highlights a shift in the strategy of theoretical cosmology. For many years, the dominant approach to model building has been bottom-up: one begins with observational anomalies or unexplained phenomena, proposes new fields or interactions capable of addressing them, and investigates whether the resulting model is compatible with existing data [35, 38, 39]. While this methodology has produced a rich landscape of candidate theories, it often postpones questions about ultraviolet consistency.

The asymptotic-safety framework reverses this order of reasoning [13, 17, 18]. Instead of beginning with phenomenology and asking how it might be embedded into a fundamental theory, one begins with the requirement that the theory remain well-defined at arbitrarily high energies. This requirement restricts the renormalization group trajectories that are physically admissible. Cosmological models must then be constructed within the subset of effective theories compatible with those trajectories.

From this perspective, ultraviolet completion becomes a primary criterion for theoretical viability rather than a secondary consideration. The case studies explored in this essay illustrate this shift concretely. The shift-symmetric Horndeski models that have been widely studied as candidates for dynamical dark energy rely on a coupling that is driven to zero by the ultraviolet fixed point [65]. The minimal Abelian Higgs model predicts cosmic strings with tensions far above the observational bounds once it is embedded into the asymptotically safe framework [47]. These results do not imply that dynamical dark energy or cosmic strings are impossible; they indicate that their realization within a UV-complete theory requires additional structure absent from the simplest models. The renormalization group therefore acts as a guide for model building by indicating which directions in theory space are compatible with a fundamental completion, and by doing so it connects the abstract requirement of quantum consistency to the concrete observational program of precision cosmology [40, 42, 60].

18 Beta Functions of the Abelian Higgs Model under Gravity

18.1 Setup and Field Content

The Abelian Higgs model coupled to quantum gravity provides the simplest arena in which the competition between gravitational antiscreening and matter-sector dynamics can be studied with full quantitative control. The minimal field content consists of a $U(1)$ gauge field A_μ, a complex charged scalar ϕ, and the metric $g_{\mu\nu}$ in the Einstein-Hilbert truncation. The FRG effective action is [47]

$$\Gamma_k = \int d^4x\sqrt{g}\left[-\frac{M_{\rm Pl}^2}{2}(R-2\Lambda) + \frac{1}{4}F_{\mu\nu}F^{\mu\nu} + |D_\mu\phi|^2 + m^2|\phi|^2 + \lambda_4|\phi|^4\right], \quad (28)$$

where $D_\mu = \partial_\mu - ig_{\rm g}A_\mu$ is the gauge-covariant derivative. In terms of dimensionless couplings $g = Gk^2$, $\lambda = \Lambda/k^2$, and rescaled matter couplings $\tilde{m}^2 = m^2/k^2$, $\tilde{\lambda}_4 = \lambda_4$, $\tilde{g}_{\rm g}^2 = g_{\rm g}^2/k^2$, the Wetterich equation generates beta functions that govern the RG flow of all five parameters.

18.2 Gauge Coupling Beta Function

The running of the gauge coupling receives contributions from the gauge boson and scalar loops (matter sector) and from the graviton loop (gravitational sector). After projecting the flow equation onto the $F_{\mu\nu}F^{\mu\nu}$ operator and evaluating the functional traces with the Litim regulator, the beta function for the dimensionless gauge coupling squared $\tilde{g}_{\rm g}^2$ takes the form [47, 67]

$$\beta_{\tilde{g}_{\rm g}^2} = \tilde{g}_{\rm g}^2\left[-2 + \frac{5G}{18\pi(1-2\Lambda)^2} - \frac{5G}{9\pi(1-2\Lambda)} + \frac{1}{48\pi^2}\,\tilde{g}_{\rm g}^2\right]. \quad (29)$$

The term -2 arises from canonical scaling ($[\tilde{g}_{\rm g}^2] = -2$ in four dimensions). The terms proportional to G represent gravitational antiscreening: they are positive and grow as $\Lambda \to 1/2$, providing the mechanism that can save the theory from the Landau pole. The final term $\propto \tilde{g}_{\rm g}^4$ is the standard matter-sector screening contribution from the charged scalar loop.

18.3 Fixed-Point Topology and the Landau-Pole Resolution

Setting $\beta_{\tilde{g}_{\rm g}^2} = 0$ and ignoring the matter sector term for simplicity, the non-trivial fixed point is located at

$$\tilde{g}_{{\rm g},*}^2 = \frac{(1+\tilde{m}^2)^2}{\sqrt{3}}\sqrt{40\pi G}, \quad (30)$$

where the scalar mass contribution enters through the propagator structure of the heat-kernel trace. This fixed point exists for any $G > 0$, confirming that gravitational antiscreening resolves the Abelian triviality problem: the theory no longer hits a Landau pole but instead approaches a finite interacting fixed point [67, 47].

The topology of the fixed-point landscape in the $(G, \tilde{g}_{\mathrm{g}}^2)$ plane is therefore qualitatively different from the flat-spacetime case. In the absence of gravity, the only fixed point of the gauge coupling is the Gaussian one at $\tilde{g}_{\mathrm{g},*}^2 = 0$, which is UV-attractive but corresponds to a trivial, non-interacting theory. With gravity, an interacting fixed point emerges at finite $\tilde{g}_{\mathrm{g},*}^2 \propto \sqrt{G}$, providing a UV completion for the interacting Abelian gauge theory.

18.4 Scalar Sector Beta Functions

The beta functions for the scalar mass and quartic coupling are

$$\beta_{\tilde{m}^2} = (-2 + \eta_\phi)\, \tilde{m}^2 + \frac{3\tilde{\lambda}_4}{16\pi^2(1+\tilde{m}^2)^2} - \frac{3\tilde{g}_{\mathrm{g}}^2}{16\pi^2(1+\tilde{m}^2)} + f_m(G, \Lambda), \tag{31}$$

$$\beta_{\tilde{\lambda}_4} = (\eta_\phi)\, \tilde{\lambda}_4 - \frac{9\tilde{\lambda}_4^2}{16\pi^2(1+\tilde{m}^2)^3} + \frac{\tilde{g}_{\mathrm{g}}^4}{4\pi^2(1+\tilde{m}^2)^2} + f_\lambda(G, \Lambda), \tag{32}$$

where η_ϕ is the scalar anomalous dimension and f_m, f_λ encode the gravitational contributions from the transverse-traceless and trace scalar graviton modes. The gauge coupling appears in equation (31) with a negative sign, reflecting the antiscreening of the mass by gauge boson fluctuations, and in equation (32) with a positive sign, reflecting the quartic coupling generated by box diagrams at one loop.

The central difficulty for cosmic string production is that the gravitational contribution $f_\lambda(G, \Lambda)$ tends to drive $\tilde{\lambda}_{4,*}$ negative, destabilizing the scalar potential. Spontaneous symmetry breaking requires a negative mass squared $\tilde{m}_*^2 < 0$ at the fixed point simultaneously with a positive quartic coupling $\tilde{\lambda}_{4,*} > 0$ to ensure boundedness from below. The minimal Abelian Higgs model coupled to gravity generically fails to satisfy both conditions simultaneously [47].

19 The Yukawa-Abelian-Higgs System and Fermionic Screening

19.1 Extension to the Yukawa Sector

The obstruction identified in Section 18 motivates the inclusion of charged Dirac fermions ψ_i $(i = 1, \ldots, N_f)$ coupled to the scalar through Yukawa interactions.

The interaction Lagrangian takes the form [47]

$$\mathcal{L}_{\text{Yuk}} = i\bar{y}_k \left(\phi^* \bar{\psi}^c \psi + \phi \bar{\psi} \psi^c\right), \tag{33}$$

where $\psi^c = C\bar{\psi}^T$ denotes the charge-conjugate spinor and $\bar{y}_k$ is the dimensionful Yukawa coupling. In terms of the dimensionless Yukawa coupling $\tilde{y}^2 = \bar{y}^2/k^2$, the extended system has beta functions for all couplings modified by fermionic loop contributions.

19.2 Modified Gauge Coupling Beta Function

With N_f Dirac fermions, the gauge coupling beta function becomes

$$\beta_{\tilde{g}_{\text{g}}^2} = \tilde{g}_{\text{g}}^2 \left[-2 + \frac{5G}{18\pi(1-2\Lambda)^2} - \frac{5G}{9\pi(1-2\Lambda)} + \frac{1+N_f}{48\pi^2}\,\tilde{g}_{\text{g}}^2\right]. \tag{34}$$

The fermion contribution enters the matter-sector term with coefficient $(1 + N_f)$: the scalar contributes 1 and each Dirac fermion contributes 1 to the screening of the gauge coupling. The fixed-point value of the gauge coupling therefore decreases as N_f increases:

$$\tilde{g}_{\text{g},*}^2 \propto \frac{1}{1+N_f}. \tag{35}$$

This screening is the mechanism by which fermions shift the locus of spontaneous symmetry breaking to lower energies. A smaller $\tilde{g}_{\text{g},*}^2$ means that the gauge boson contribution to the scalar mass beta function (31) is reduced, allowing $\tilde{m}_*^2 < 0$ to be achieved at a sub-Planckian scale.

19.3 Yukawa Coupling Beta Function

The Yukawa coupling itself evolves under the RG according to

$$\beta_{\tilde{y}^2} = -2\tilde{y}^2 + \frac{(4+N_f)\tilde{y}^4}{8\pi^2} - \frac{3\tilde{g}_{\text{g}}^2\,\tilde{y}^2}{8\pi^2} + f_y(G,\Lambda), \tag{36}$$

where the canonical scaling term $-2\tilde{y}^2$ reflects the mass dimension -1 of the Yukawa coupling in four dimensions, and $f_y(G,\Lambda)$ contains gravitational contributions. A non-trivial fixed point $\tilde{y}_*^2 > 0$ exists when the balance between the positive $(4 + N_f)\tilde{y}^4$ term and the gauge-coupling screening determines the fixed-point value.

The ratio $\tilde{g}_{\text{g},*}^2/\tilde{y}_*^2$ plays the critical role in determining whether the scalar potential admits spontaneous symmetry breaking. Specifically, the fixed-point condition $\tilde{m}_*^2 < 0$ is satisfied only when $\tilde{g}_{\text{g},*}^2/\tilde{y}_*^2$ is sufficiently small, which requires

N_f large enough to screen the gauge coupling below the Yukawa coupling [47].

19.4 Quantitative Stability Thresholds

The detailed stability analysis of the coupled fixed-point equations reveals two distinct thresholds:

For $N_f < 22$, no parameter region in the (G, Λ) plane admits simultaneous solutions $\tilde{m}_*^2 < 0$ and $\tilde{\lambda}_{4,*} > 0$. The scalar potential is universally unstable: gauge boson fluctuations drive the quartic coupling negative before the mass can become negative, producing a potential unbounded from below.

For $22 \leq N_f < 42$, positive solutions for $\tilde{\lambda}_{4,*}$ exist but only within a restricted "strip" in the (Λ, G) plane at $\Lambda < 0$. The allowed region is narrow and disappears as $\Lambda \to 0$, indicating that cosmologically viable symmetry breaking requires a large negative cosmological constant at the Planck scale.

For $N_f \geq 42$, the stability conditions are satisfied across a broad range of the cosmological constant including a neighborhood of $\Lambda = 0$. The cosmological constant sensitivity becomes $|\Lambda_{\text{threshold}}| \lesssim 7.2$ for $N_f = 42$, improving as N_f increases [47]. This threshold at $N_f = 42$ constitutes the complexity tax quantified in Section 5.3.

20 Explicit Beta Functions for Shift-Symmetric Horndeski Couplings

20.1 The Complete Beta Functions at Leading Order

Within the shift-symmetric Horndeski truncation, the beta functions for the dimensionless braiding coupling h and the quartic coupling g can be obtained from the $P^{-1}F$ expansion of the Wetterich equation, evaluated on a maximally symmetric background with the York decomposition of metric fluctuations and the Litim regulator [65]. To leading order in the gravitational couplings, the results are

$$\beta_h = \left[3 + \frac{g}{4\pi^2} + G\left(\frac{7}{4\pi(3-4\Lambda)} + \frac{9}{2\pi(3-4\Lambda)^2} - \frac{5}{9\pi(1-2\Lambda)} - \frac{20}{9\pi(1-2\Lambda)^2}\right)\right] h, \tag{37}$$

$$\beta_g = 4g + \frac{9}{32\pi^2} g^2 + \frac{Gg}{6\pi}\left[\frac{8}{3-4\Lambda} + \frac{15}{(3-4\Lambda)^2} - \frac{20}{(1-2\Lambda)^2}\right] + \mathcal{O}(G^2). \tag{38}$$

20.2 Structure and Interpretation of β_h

The beta function (37) has a definitive structure: h appears as an overall factor on the right-hand side, confirming the symmetry argument that $h_* = 0$ is an exact fixed point. The bracket contains three types of contribution. The term $+3$ is the canonical scaling, arising from the mass dimension $[h] = -3$ and the rescaling $h = k^{-3}\tilde{h}$. The term $g/(4\pi^2)$ is the matter self-interaction correction from the χ^2 vertex appearing in the loop diagram at order g. The remaining gravitational terms arise from the transverse-traceless graviton loops (proportional to the $(3 - 4\Lambda)$ denominators) and the trace scalar graviton loops (proportional to the $(1 - 2\Lambda)$ denominators).

The sign of the gravitational bracket is crucial. Expanding near $G = 0$ and $\Lambda = 0$, one finds that the gravitational contribution is negative, meaning that gravity reduces the canonical suppression of h but cannot change the sign of the bracket without exceeding the weak-gravity bound $G \lesssim 1.4$ [65, 18]. The critical exponent θ_h at the SGFP is given by minus the value of the bracket at $g = g_*$, $G = G_*$, $\Lambda = \Lambda_*$:

$$\theta_h = -\beta_h/h\big|_{\text{SGFP}} = -\left[3 + \frac{g_*}{4\pi^2} + (\text{gravitational terms})\right]. \tag{39}$$

Since the gravitational terms cannot make the bracket negative without violating the weak-gravity bound, $\theta_h < 0$ throughout the accessible region of the (G, Λ) plane. The braiding coupling is therefore irrelevant at all physically viable gravitational fixed points.

20.3 Structure and Interpretation of β_g

The beta function (38) has a qualitatively different structure. The canonical scaling term is $+4g$, reflecting the mass dimension $[g] = -4$. The matter sector correction $9g^2/(32\pi^2)$ is the standard quartic self-interaction. The gravitational contribution at order G does not vanish at $g = 0$: it is proportional to Gg, meaning that it shifts the fixed-point value from zero but does not introduce a g-independent term that would produce a non-zero fixed point from the Gaussian starting value $g = 0$ through a pure gravitational mechanism.

The fixed-point condition $\beta_g = 0$ gives

$$g_* = -\frac{4 + f_G(G_*, \Lambda_*)}{9/(32\pi^2)}, \tag{40}$$

where f_G denotes the gravitational bracket in equation (38) evaluated at the fixed point. For the gravitational couplings at the NGFP, $G_* \approx 0.7$–1.2 and $\Lambda_* \approx 0.1$–0.2 (truncation dependent), one finds $g_* < 0$ at the SGFP. The quartic coupling

is therefore shifted to a small negative value by gravity, which is not the large positive value $g_{\text{required}} \approx 9 \times 10^7\,\text{eV}^{-4}$ required for dark energy [65].

21 Fixed-Point Analysis and the Stability Matrix

21.1 The Stability Matrix at the SGFP

The stability properties of the shifted Gaussian fixed point are encoded in the stability matrix $M_{ij} = \partial\beta_{g_i}/\partial g_j|_{g_*}$ evaluated at the fixed-point values $(h_*, g_*, G_*, \Lambda_*)$. Since $h_* = 0$ and the beta functions factorize as $\beta_h = h \cdot f(g, G, \Lambda)$, the h-direction decouples from the (g, G, Λ) sector at the linearized level. The stability matrix therefore has a block-diagonal structure:

$$M = \begin{pmatrix} M_h & 0 \\ 0 & M_{g,G,\Lambda} \end{pmatrix}, \tag{41}$$

where $M_h = \partial\beta_h/\partial h|_{h_*=0} = f(g_*, G_*, \Lambda_*)$ is a single scalar entry and $M_{g,G,\Lambda}$ is a 3×3 matrix for the remaining couplings.

21.2 Critical Exponents

The critical exponents are the negatives of the eigenvalues of M: $\theta_I = -\text{eig}(M)$. The critical exponent associated with h is

$$\theta_h = -M_h = -\left[3 + \frac{g_*}{4\pi^2} + (\text{grav. bracket})\right]. \tag{42}$$

For representative values $G_* = 1.0$, $\Lambda_* = 0.1$, $g_* \approx -0.02$ (SGFP), the gravitational bracket in equation (37) evaluates to approximately -0.8, giving

$$\theta_h \approx -(3 - 0.0005 - 0.8) \approx -2.2. \tag{43}$$

This negative value confirms the irrelevance of h at the SGFP. The canonical scaling $+3$ dominates over the gravitational correction -0.8, leaving a net negative critical exponent of order two [65].

The RG trajectory emerging from the SGFP therefore has $h(k) \propto (k/M_{\text{Pl}})^{|\theta_h|} \approx (k/M_{\text{Pl}})^{2.2}$ as the scale k decreases from the Planck scale. At the cosmological scale $k \sim H_0$,

$$h(H_0) \sim \left(\frac{H_0}{M_{\text{Pl}}}\right)^{2.2} \approx \left(10^{-61}\right)^{2.2} \approx 10^{-134}. \tag{44}$$

This is a fantastically small number. The observationally required value is $h \sim H_0^{-3} M_{\text{Pl}}^{-0} \approx 10^{126}\,\text{eV}^{-3}$, implying a mismatch of roughly 260 orders of magnitude

in the dimensionful coupling $\bar{h} = hk^3$ evaluated at $k = H_0$ [65].

21.3 Near-Critical Regime and the Weak-Gravity Bound

Near the weak-gravity bound $G \to G_{\text{crit}} \approx 1.4$ (at $\Lambda = 0$), the gravitational bracket in equation (37) increases in magnitude. One can ask whether the bracket can become large enough to produce $\theta_h > 0$, which would make h relevant and potentially allow non-zero braiding in the IR. A systematic scan of the accessible (G, Λ) region shows that this does not occur: the bracket reaches at most approximately -1.5 before the matter fixed points disappear at the weak-gravity bound, giving $\theta_h \approx -(3 - 1.5) = -1.5$ rather than zero [65, 18]. The braiding coupling remains irrelevant throughout the accessible parameter space.

22 The EFT-AS Gap and the Cosmological Matching Problem

22.1 Effective Field Theory Requirements for Dark Energy

From the effective field theory perspective, the cosmological constant problem requires that the quartic coupling g and braiding coupling h take specific infrared values to reproduce the observed dark energy density $\rho_\Lambda \approx (2.3 \times 10^{-3}\,\text{eV})^4 \approx 2.8 \times 10^{-47}\,\text{GeV}^4$. In the shift-symmetric Horndeski model with $h \neq 0$, the scalar field energy density on a de Sitter background evaluates to contributions from both g and h, requiring [65]

$$g_{\text{EFT}} \approx \frac{\rho_\Lambda}{H_0^4} \approx \frac{10^{-47}\,\text{GeV}^4}{(2.3 \times 10^{-42}\,\text{GeV})^4} \approx 9 \times 10^7\,\text{eV}^{-4}, \tag{45}$$

$$h_{\text{EFT}} \approx \frac{1}{H_0^3\, M_{\text{Pl}}^0} \approx \frac{1}{(2.3 \times 10^{-42}\,\text{GeV})^3} \approx 2.3 \times 10^{33}\,\text{eV}^{-3}. \tag{46}$$

These are enormous numbers in natural units. They correspond to the intrinsic scale of the cosmological constant fine-tuning problem, expressed now as specific requirements on Horndeski coupling values.

22.2 Asymptotic-Safety Predictions versus EFT Requirements

The asymptotic-safety prediction for the RG-evolved coupling g at the cosmological scale is obtained by running the quartic coupling from the Planck scale $k = M_{\text{Pl}}$

to the Hubble scale $k = H_0$. Using the linearized solution near the SGFP:

$$g(H_0) \approx g_* + C_g \left(\frac{H_0}{M_{\rm Pl}}\right)^{|\theta_g|} \approx g_* + C_g \times 10^{-244}. \tag{47}$$

The integration constant C_g is of order one for trajectories on the UV-critical surface. The running therefore produces $g(H_0) \approx g_* \approx -0.02$, in natural units where the RG scale sets the mass scale, which translates to the dimensionful coupling $\bar{g}(H_0) = g_* H_0^4 \approx -0.02 \times (2.3 \times 10^{-42}\,\text{GeV})^4 \approx -6 \times 10^{-49}\,\text{GeV}^4$.

Converting to the observational units used in the EFT comparison:

$$g_{\rm AS}(H_0) \approx g_* \times H_0^{-4} \approx -0.02 \times (10^{-42}\,\text{GeV})^{-4} \approx -2 \times 10^{79}\,\text{GeV}^{-4} \approx -2 \times 10^{-7}\,\text{eV}^{-4}. \tag{48}$$

Comparing with $g_{\rm EFT} \approx 9 \times 10^7\,\text{eV}^{-4}$, the mismatch is a factor of $\sim 10^{15}$. The sign is also wrong: asymptotic safety predicts a small negative g_*, while the EFT requirement is a large positive $g_{\rm EFT}$ [65].

For the braiding coupling the situation is even starker. Asymptotic safety predicts $h(H_0) \approx 0$ to all practical purposes, while the EFT requirement is $h_{\rm EFT} \approx 2.3 \times 10^{33}\,\text{eV}^{-3}$.

22.3 Implications for the Hubble Tension Resolution

The EFT-AS gap has direct implications for proposed resolutions of the Hubble tension. Models in which the Horndeski braiding parameter $\alpha_B \neq 0$ generates a time-varying dark energy equation of state can shift the CMB-inferred H_0 toward the locally measured value. However, such models require $h_{\rm EFT} \sim 10^{33}\,\text{eV}^{-3}$, whereas asymptotic safety predicts $h(H_0) \sim 10^{-134}$ in the same units—a discrepancy of 167 orders of magnitude. No modification to the polynomial truncation or to the regulator scheme is capable of bridging a gap of this magnitude [65, 49]. The Hubble tension cannot be resolved through shift-symmetric kinetic braiding in a UV-complete theory.

The σ_8 tension, which involves the growth rate of structure, faces the same obstruction: modifications to the gravitational slip parameter through $\alpha_B \neq 0$ require the same non-zero braiding coupling that asymptotic safety forces to zero. Both cosmological tensions therefore remain unaddressed by the simplest asymptotically safe Horndeski models [40, 41].

23 Gravitational Wave Signatures and the Observational Bridge

23.1 Cosmic String Networks and the Stochastic Background

A cosmic string network formed at a symmetry-breaking scale v generates a stochastic gravitational-wave background through the oscillation and decay of string loops. The energy spectrum of this background, expressed as the fractional energy density per logarithmic frequency interval, takes the form [42, 44]

$$\Omega_{\rm GW}(f) = \frac{8\pi}{3H_0^2} f \, \frac{dE_{\rm GW}}{dV\, df}, \tag{49}$$

where the frequency-dependent emission rate depends on the string tension $\mu = 2\pi v^2 \ln(\lambda_4/g_{\rm g}^2)^{1/2}$, the loop number density, and the loop lifetime. In the relevant frequency band of pulsar timing arrays $f \sim 1\,{\rm yr}^{-1} \approx 3 \times 10^{-8}\,{\rm Hz}$, the spectrum of a network of Nambu-Goto strings in the scaling regime scales approximately as a power law

$$\Omega_{\rm GW}(f) \propto (G\mu)^x \, f^n, \tag{50}$$

where $G\mu$ is the dimensionless string tension (Newton's constant times string tension), $x \approx 1$, and n depends on the loop distribution model [42].

The NANOGrav 15-year dataset reports a detection of the stochastic background with amplitude and spectral index [42]

$$h_c(f) \approx A_{\rm GW} \left(\frac{f}{f_{\rm yr}}\right)^{(3-\gamma)/2}, \quad A_{\rm GW} \approx 2.4 \times 10^{-15}, \quad \gamma \approx 3.2, \tag{51}$$

where $f_{\rm yr} = 1\,{\rm yr}^{-1}$. The Hellings-Downs angular correlation function, characteristic of a gravitational-wave background, was detected at $\sim 4\sigma$ significance. However, the spectral shape and amplitude are consistent with a population of supermassive black-hole binaries and inconsistent with the spectrum expected from a stable Nambu-Goto string network at the $v \lesssim 4.6 \times 10^{14}\,{\rm GeV}$ scale permitted by observational bounds [42, 47].

23.2 Connecting String Tension to the Symmetry-Breaking Scale

The constraint from NANOGrav on the symmetry-breaking scale can be derived by requiring that the theoretical prediction for $\Omega_{\rm GW}$ not exceed the observed signal at the relevant frequencies. For a stable cosmic string network the constraint

translates to an upper bound on the string tension

$$G\mu \lesssim 2 \times 10^{-10}, \tag{52}$$

which corresponds to a symmetry-breaking scale

$$v \lesssim 4.6 \times 10^{14}\,\text{GeV} \approx 3.8 \times 10^{-5} M_{\text{Pl}}. \tag{53}$$

The asymptotic-safety prediction for the minimal Abelian Higgs model is $v \sim M_{\text{Pl}} = 1.22 \times 10^{19}\,\text{GeV}$ [47]. The ratio of the predicted and allowed scales is

$$\frac{v_{\text{AS}}}{v_{\text{max}}} \approx \frac{M_{\text{Pl}}}{3.8 \times 10^{-5} M_{\text{Pl}}} \approx 2.6 \times 10^{4}. \tag{54}$$

This is the scale mismatch that the forty-two fermion extension must address: the fermions must screen the gauge coupling enough to shift v downward by over four orders of magnitude from the Planck scale.

23.3 The AS-GW Interface as a Precision Test

The chain of reasoning connecting asymptotic safety to the gravitational wave background is a closed logical loop: UV fixed-point structure constrains v through the gauge coupling fixed point, v determines $G\mu$, $G\mu$ determines the amplitude and spectral shape of $\Omega_{\text{GW}}(f)$, and $\Omega_{\text{GW}}(f)$ is constrained by NANOGrav. As the sensitivity of pulsar timing arrays improves—with the Square Kilometre Array (SKA) expected to reach $G\mu \sim 10^{-12}$—the constraint (53) will tighten, further probing the parameter space of UV-complete cosmic string models [47, 60]. Either a signal consistent with a sub-Planckian string scale is detected, requiring the forty-two fermion extension to be part of the true theory, or no string signal is found, consistent with the minimal AS prediction that cosmic strings are a Planck-scale phenomenon invisible to current and near-future PTAs.

23.4 Tensor Modes and the Speed of Gravitational Waves

The GW170817 constraint on the tensor propagation speed, $|c_t - 1| \lesssim 10^{-15}$ [43], provides a separate and complementary constraint on the Horndeski sector. Within the Horndeski framework the tensor speed is [35, 34]

$$c_t^2 = 1 + \frac{2XG_{4,X}}{M_{\text{Pl}}^2 + 2XG_{4,X}}, \tag{55}$$

where $G_{4,X} = \partial G_4/\partial X$ is the derivative of the L_4 Horndeski function with respect to the kinetic variable $X = -\chi$. The GW170817 constraint requires $G_{4,X} \approx 0$, which eliminates the L_4 and L_5 sectors and restricts viable Horndeski theories to the L_2-L_3 subspace [65]. This is precisely the shift-symmetric kinetic braiding sector analyzed in this work. The constraint $c_t = 1$ is therefore not an independent result but a consistency condition that uniquely selects the sector in which the asymptotic-safety analysis applies.

24 Conclusion

The asymptotic-safety paradigm, implemented through the Functional Renormalization Group, provides a powerful top-down filter on the space of phenomenological models for physics beyond the Standard Model. We have reviewed in detail how this filter operates for two model classes of direct observational relevance: shift-symmetric Horndeski theories with kinetic braiding, and the Abelian Higgs model as a source of cosmic strings.

For kinetic braiding, the conclusion is definitive within the polynomial truncation: the braiding coupling h is symmetry-protected and vanishes identically at the shifted Gaussian fixed point. Its IR value is predicted to be zero, which is incompatible with the non-zero braiding required for dynamical dark energy [65]. The quartic coupling g is similarly suppressed by approximately fifty-seven orders of magnitude relative to the value needed to match the observed dark energy density. Kinetic braiding models are swampland candidates: they cannot be connected to an asymptotically safe UV completion without abandoning the phenomenological features that motivate them [49].

For cosmic strings in the Abelian Higgs model, the conclusion is more conditional: the minimal model fails, but an extended model with at least forty-two charged Dirac fermions may satisfy the UV consistency conditions [47]. This forty-two fermion barrier quantifies the complexity tax imposed by asymptotic safety on the cosmic string hypothesis. The recent NANOGrav results [42], which disfavor a stable-string interpretation of the observed stochastic gravitational-wave background in favor of supermassive black-hole binaries, provide independent observational support for the theoretical exclusion of the minimal model.

These results collectively support a picture of the asymptotically safe landscape as a small landscape: the UV consistency requirement reduces the apparently vast space of possible low-energy effective theories to a much smaller set of genuinely viable candidates [17, 16, 60]. This reduction is not a limitation of the framework but its defining virtue. A theory that cannot explain the origin of its own parameters from first principles is not a fundamental theory; it is an approx-

imation [2, 13]. The asymptotic-safety program, by demanding that every coupling either flow to a fixed-point value or be genuinely measurable, transforms the question of model selection from a matter of phenomenological preference into a matter of mathematical necessity.

The program is not complete. Non-polynomial truncations, Lorentzian extensions, and more general geometric frameworks remain open directions that may modify or extend the conclusions reviewed here [54, 50]. But the methodology is established, the tools are in place, and the first generation of results is already shaping the landscape of viable beyond-Standard-Model physics in ways that decades of bottom-up phenomenology alone could not achieve [18, 45]. The universe we observe may ultimately be far more constrained—and far more elegant—than the space of effective theories would suggest.

Appendices

A Derivation of the Wetterich Flow Equation

The central object of the Functional Renormalization Group is the scale-dependent effective action Γ_k. Its flow with respect to the infrared cutoff scale k is governed by the Wetterich equation [9].

We begin with the partition function in the presence of a regulator term that suppresses infrared modes:

$$Z_k[J] = \int \mathcal{D}\Phi \ \exp(-S[\Phi] - \Delta S_k[\Phi] + J \cdot \Phi)\,, \tag{56}$$

where Φ collectively denotes all fields and the regulator term is $\Delta S_k = \frac{1}{2}\Phi R_k \Phi$. The generating functional of connected correlators is $W_k[J] = \ln Z_k[J]$, and the classical field is defined by $\Phi_{\text{cl}} = \delta W_k/\delta J$. The scale-dependent effective action is obtained through a modified Legendre transform:

$$\Gamma_k[\Phi_{\text{cl}}] = J \cdot \Phi_{\text{cl}} - W_k[J] - \tfrac{1}{2}\Phi_{\text{cl}} R_k \Phi_{\text{cl}}. \tag{57}$$

From the definition of Γ_k and the stationarity condition $\delta\Gamma_k/\delta\Phi_{\text{cl}} = J - R_k\Phi_{\text{cl}}$, one obtains the second functional derivative relation

$$\left(\Gamma_k^{(2)} + R_k\right)_{xy} = \left(W_k^{(2)}\right)^{-1}_{xy}. \tag{58}$$

Differentiating the modified Legendre transform with respect to the scale parameter $t = \ln k$ and using the chain rule on both sides, one arrives at the exact Wetterich flow equation [9, 10]:

$$\partial_t \Gamma_k = \frac{1}{2}\,\text{Tr}\left[(\Gamma_k^{(2)} + R_k)^{-1}\partial_t R_k\right]. \tag{59}$$

This equation has the structural form of a one-loop expression but is exact: all quantum corrections are encoded in the full propagator $(\Gamma_k^{(2)} + R_k)^{-1}$, which itself depends on Γ_k [12, 11].

B York Decomposition of Metric Fluctuations

In functional renormalization group calculations for gravity, it is convenient to decompose metric fluctuations into irreducible components under diffeomorphisms [14, 16].

The background-field method writes the full metric as $g_{\mu\nu} = \bar{g}_{\mu\nu} + h_{\mu\nu}$. The York decomposition expresses $h_{\mu\nu}$ as

$$h_{\mu\nu} = h^{\rm TT}_{\mu\nu} + \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu + \nabla_\mu \nabla_\nu \sigma - \tfrac{1}{4}\bar{g}_{\mu\nu}\nabla^2\sigma + \tfrac{1}{4}\bar{g}_{\mu\nu} h, \tag{60}$$

where the transverse-traceless (TT) modes satisfy

$$\nabla^\mu h^{\rm TT}_{\mu\nu} = 0, \qquad \bar{g}^{\mu\nu} h^{\rm TT}_{\mu\nu} = 0. \tag{61}$$

The decomposition therefore separates the metric fluctuations into spin-2 transverse-traceless modes $h^{\rm TT}_{\mu\nu}$ carrying the propagating graviton degrees of freedom, transverse vector modes ξ_μ, and two scalar modes σ and h. In the Landau-DeWitt gauge the vector and longitudinal scalar modes decouple from the physical spectrum. The TT sector typically dominates the RG flow of the gravitational couplings, while the trace scalar mode couples to matter fields through non-minimal interaction operators such as $\zeta\chi R$ [65].

C Heat-Kernel Evaluation of Functional Traces

Functional traces appearing in the Wetterich equation can be evaluated using heat-kernel techniques [10, 12]. Consider an operator of Laplace type $\Delta = -\nabla^2 + U$. The trace of a function of this operator can be written as

$$\mathrm{Tr}\, f(\Delta) = \int d^4x \sqrt{g} \sum_{n=0}^{\infty} Q_{4-n}[f]\, a_n(\Delta), \tag{62}$$

where the a_n are the Seeley-DeWitt heat-kernel coefficients, and the Q-functionals are defined by

$$Q_n[f] = \frac{1}{\Gamma(n)} \int_0^\infty ds\, s^{n-1} f(s). \tag{63}$$

The leading coefficients are $a_0 = 1/(4\pi)^2$, $a_2 = R/(6(4\pi)^2)$, and higher terms involving curvature invariants. For the optimized Litim regulator $R_k(p^2) = (k^2 - p^2)\Theta(k^2 - p^2)$, the Q-functionals reduce to simple rational expressions:

$$Q_n\left[\frac{\partial_t R_k}{\Gamma_k^{(2)} + R_k}\right] = \frac{2k^{2n}}{(k^2 + m^2)^p}, \tag{64}$$

for appropriate powers p depending on n and the regulator form. Using these identities, the functional trace in the Wetterich equation can be expressed as a polynomial in the curvature invariants and field-dependent masses, allowing the beta functions of gravitational and matter couplings to be extracted systematically

[11, 10].

D Projection of Beta Functions

To determine the renormalization group flow of specific couplings, one projects the flow equation onto the corresponding operators [11, 12]. Consider a truncated effective action

$$\Gamma_k = \sum_i g_i(k) \int d^4x \sqrt{g}\, \mathcal{O}_i. \tag{65}$$

Applying the flow equation yields $\partial_t \Gamma_k = \frac{1}{2}\,\mathrm{Tr}[(\Gamma_k^{(2)} + R_k)^{-1}\partial_t R_k]$. To obtain the beta function for a coupling g_i, one isolates the coefficient of the operator $\mathcal{O}_i$:

$$\beta_{g_i} = \frac{\partial}{\partial \mathcal{O}_i}\left(\partial_t \Gamma_k\right). \tag{66}$$

In practice this projection is implemented by evaluating the flow on specific background field configurations. For example, to extract the flow of the Einstein-Hilbert term, one evaluates the trace on a maximally symmetric background where

$$R_{\mu\nu\rho\sigma} = \frac{R}{12}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}). \tag{67}$$

Scalar-sector couplings are obtained by expanding around constant scalar-field backgrounds and identifying the coefficients of the corresponding operators. For the Horndeski system, the beta functions β_h and β_g are extracted by projecting onto terms proportional to $\chi\Box\phi$ and χ^2 respectively, using the $P^{-1}F$ expansion described in Section 7 [65].

E Renormalization Group Trajectories Near the Fixed Point

Close to the ultraviolet fixed point, the RG flow is governed by the linearized equation (see Section 3.1)

$$k\partial_k \delta g_i = B_{ij}\, \delta g_j, \tag{68}$$

where $B_{ij} = \partial\beta_i/\partial g_j|_{g_*}$ is the stability matrix and $\delta g_i = g_i - g_i^*$ denotes the deviation from the fixed-point values. Diagonalizing the matrix B yields eigenvectors v_I and eigenvalues $-\theta_I$. The general solution is

$$g_i(k) = g_i^* + \sum_I C_I v_I^{(i)} \left(\frac{k}{k_0}\right)^{-\theta_I}, \tag{69}$$

where k_0 is a reference scale and C_I are integration constants [2, 15]. The constants C_I associated with relevant directions ($\theta_I > 0$) are the free parameters of the theory, to be fixed by experiment; those associated with irrelevant directions ($\theta_I < 0$) are driven to zero as $k \to \infty$ and are therefore predicted by the UV fixed point.

For the shifted Gaussian fixed point of the Horndeski system, the critical exponent θ_h associated with the braiding coupling satisfies $\theta_h < 0$, confirming the irrelevance of h. As k decreases from the Planck scale toward cosmological scales, $h(k)$ remains exponentially suppressed:

$$h(k) \sim h_* + C_h \left(\frac{k}{M_{\rm Pl}}\right)^{|\theta_h|}, \tag{70}$$

with $h_* = 0$. The infrared value $h(H_0)$ is therefore suppressed by the ratio $(H_0/M_{\rm Pl})^{|\theta_h|} \approx 10^{-60|\theta_h|}$, making it completely negligible for cosmological phenomenology [65].

F Operator Dimension Table and Truncation Systematics

The operators entering the FRG truncation can be organized systematically according to their canonical mass dimension $[\mathcal{O}]$ in four Euclidean dimensions [2, 7]. For a d-dimensional spacetime integral $\int d^4x \sqrt{g}\, \mathcal{O}$, the action has dimension zero, so the coupling constant $g_{\mathcal{O}}$ has dimension $4-[\mathcal{O}]$. Operators with $[\mathcal{O}] > 4$ are power-counting irrelevant.

For the scalar field in four dimensions, the canonical dimension is $[\phi] = 1$, giving $[\chi] = [\partial\phi\, \partial\phi] = 4$. The coupling to the standard kinetic term $\int \sqrt{g}\, \chi$ is therefore dimensionless. The cubic operator $\chi\Box\phi$ has dimension $[\chi\Box\phi] = 4 + 1 + 2 = 7$, giving $[h] = 4 - 7 = -3$. The quartic operator χ^2 has dimension $[\chi^2] = 8$, giving $[g] = 4 - 8 = -4$. These values confirm the power-counting irrelevance of both operators in the absence of gravity.

The gravitational sector introduces the Newton coupling $G_N = M_{\rm Pl}^{-2}$ with dimension $[G_N] = -2$ in four dimensions, or equivalently the dimensionless combination $g = G_N k^2$ used throughout this essay. The cosmological constant $\bar{\Lambda}$ has dimension $[\bar{\Lambda}] = 2$, corresponding to the dimensionless $\Lambda = \bar{\Lambda} k^{-2}$.

The beta functions in terms of dimensionless couplings are obtained by substituting $g_{\mathcal{O}} = k^{4-[\mathcal{O}]}\, \tilde{g}_{\mathcal{O}}$ and differentiating with respect to $t = \ln k$, yielding equation (6) with the canonical scaling term $-d_i \tilde{g}_i$ where $d_i = 4-[\mathcal{O}_i]$. For h and g, this gives canonical scaling exponents of $+3$ and $+4$ respectively—the "restoring force" that drives these couplings toward zero in the absence of quantum corrections. The gravitational antiscreening contribution $\beta_g^{\rm grav}$ must overcome this canonical scal-

ing by an amount sufficient to produce $g_* \neq 0$, which it does for any $G > 0$, while for h the symmetry argument ensures that $\beta_h^{\text{grav}} \propto h$ and cannot override the symmetry protection [65, 18].

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